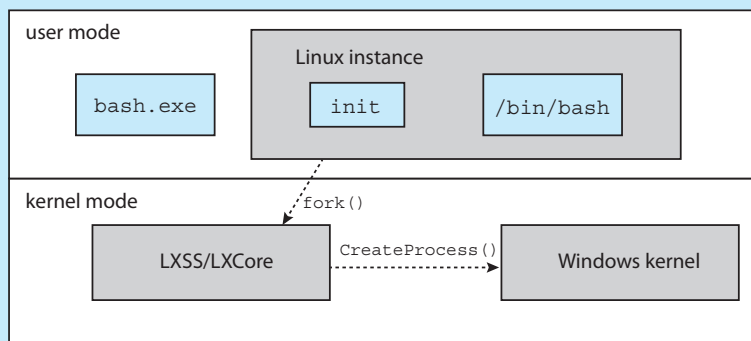


At the bottom of Android's software stack is the Linux kernel. Google has modified the Linux kernel used in Android in a variety of areas to support the special needs of mobile systems, such as power management. It has also made changes in memory management and allocation and has added a new form of IPC known as *Binder* (which we will cover in Section 3.8.2.1).

WINDOWS SUBSYSTEM FOR LINUX

Windows uses a hybrid architecture that provides subsystems to emulate different operating-system environments. These user-mode subsystems communicate with the Windows kernel to provide actual services. Windows 10 adds a Windows subsystem for Linux ([WSL](#)), which allows native Linux applications (specified as ELF binaries) to run on Windows 10. The typical operation is for a user to start the Windows application `bash.exe`, which presents the user with a `bash` shell running Linux. Internally, the WSL creates a [Linux instance](#) consisting of the `init` process, which in turn creates the `bash` shell running the native Linux application `/bin/bash`. Each of these processes runs in a Windows [Pico](#) process. This special process loads the native Linux binary into the process's own address space, thus providing an environment in which a Linux application can execute.

Pico processes communicate with the kernel services `LXCore` and `LXSS` to translate Linux system calls, if possible using native Windows system calls. When the Linux application makes a system call that has no Windows equivalent, the `LXSS` service must provide the equivalent functionality. When there is a one-to-one relationship between the Linux and Windows system calls, `LXSS` forwards the Linux system call directly to the equivalent call in the Windows kernel. In some situations, Linux and Windows have system calls that are similar but not identical. When this occurs, `LXSS` will provide some of the functionality and will invoke the similar Windows system call to provide the remainder of the functionality. The Linux `fork()` provides an illustration of this: The Windows `CreateProcess()` system call is similar to `fork()` but does not provide exactly the same functionality. When `fork()` is invoked in WSL, the `LXSS` service does some of the initial work of `fork()` and then calls `CreateProcess()` to do the remainder of the work. The figure below illustrates the basic behavior of WSL.



2.9 Building and Booting an Operating System

It is possible to design, code, and implement an operating system specifically for one specific machine configuration. More commonly, however, operating systems are designed to run on any of a class of machines with a variety of peripheral configurations.

2.9.1 Operating-System Generation

Most commonly, a computer system, when purchased, has an operating system already installed. For example, you may purchase a new laptop with Windows or macOS preinstalled. But suppose you wish to replace the preinstalled operating system or add additional operating systems. Or suppose you purchase a computer without an operating system. In these latter situations, you have a few options for placing the appropriate operating system on the computer and configuring it for use.

If you are generating (or building) an operating system from scratch, you must follow these steps:

1. Write the operating system source code (or obtain previously written source code).
2. Configure the operating system for the system on which it will run.
3. Compile the operating system.
4. Install the operating system.
5. Boot the computer and its new operating system.

Configuring the system involves specifying which features will be included, and this varies by operating system. Typically, parameters describing how the system is configured is stored in a configuration file of some type, and once this file is created, it can be used in several ways.

At one extreme, a system administrator can use it to modify a copy of the operating-system source code. Then the operating system is completely compiled (known as a **system build**). Data declarations, initializations, and constants, along with compilation, produce an output-object version of the operating system that is tailored to the system described in the configuration file.

At a slightly less tailored level, the system description can lead to the selection of precompiled object modules from an existing library. These modules are linked together to form the generated operating system. This process allows the library to contain the device drivers for all supported I/O devices, but only those needed are selected and linked into the operating system. Because the system is not recompiled, system generation is faster, but the resulting system may be overly general and may not support different hardware configurations.

At the other extreme, it is possible to construct a system that is completely modular. Here, selection occurs at execution time rather than at compile or link time. System generation involves simply setting the parameters that describe the system configuration.

The major differences among these approaches are the size and generality of the generated system and the ease of modifying it as the hardware configuration changes. For embedded systems, it is not uncommon to adopt the first approach and create an operating system for a specific, static hardware configuration. However, most modern operating systems that support desktop and laptop computers as well as mobile devices have adopted the second approach. That is, the operating system is still generated for a specific hardware configuration, but the use of techniques such as loadable kernel modules provides modular support for dynamic changes to the system.

We now illustrate how to build a Linux system from scratch, where it is typically necessary to perform the following steps:

1. Download the Linux source code from <http://www.kernel.org>.
2. Configure the kernel using the “`make menuconfig`” command. This step generates the `.config` configuration file.
3. Compile the main kernel using the “`make`” command. The `make` command compiles the kernel based on the configuration parameters identified in the `.config` file, producing the file `vmlinuz`, which is the kernel image.
4. Compile the kernel modules using the “`make modules`” command. Just as with compiling the kernel, module compilation depends on the configuration parameters specified in the `.config` file.
5. Use the command “`make modules_install`” to install the kernel modules into `vmlinuz`.
6. Install the new kernel on the system by entering the “`make install`” command.

When the system reboots, it will begin running this new operating system.

Alternatively, it is possible to modify an existing system by installing a Linux virtual machine. This will allow the host operating system (such as Windows or macOS) to run Linux. (We introduced virtualization in Section 1.7 and cover the topic more fully in Chapter 18.)

There are a few options for installing Linux as a virtual machine. One alternative is to build a virtual machine from scratch. This option is similar to building a Linux system from scratch; however, the operating system does not need to be compiled. Another approach is to use a Linux virtual machine appliance, which is an operating system that has already been built and configured. This option simply requires downloading the appliance and installing it using virtualization software such as VirtualBox or VMware. For example, to build the operating system used in the virtual machine provided with this text, the authors did the following:

1. Downloaded the Ubuntu ISO image from <https://www.ubuntu.com/>
2. Instructed the virtual machine software VirtualBox to use the ISO as the bootable medium and booted the virtual machine
3. Answered the installation questions and then installed and booted the operating system as a virtual machine

2.9.2 System Boot

After an operating system is generated, it must be made available for use by the hardware. But how does the hardware know where the kernel is or how to load that kernel? The process of starting a computer by loading the kernel is known as **booting** the system. On most systems, the boot process proceeds as follows:

1. A small piece of code known as the **bootstrap program** or **boot loader** locates the kernel.
2. The kernel is loaded into memory and started.
3. The kernel initializes hardware.
4. The root file system is mounted.

In this section, we briefly describe the boot process in more detail.

Some computer systems use a multistage boot process: When the computer is first powered on, a small boot loader located in nonvolatile firmware known as **BIOS** is run. This initial boot loader usually does nothing more than load a second boot loader, which is located at a fixed disk location called the **boot block**. The program stored in the boot block may be sophisticated enough to load the entire operating system into memory and begin its execution. More typically, it is simple code (as it must fit in a single disk block) and knows only the address on disk and the length of the remainder of the bootstrap program.

Many recent computer systems have replaced the BIOS-based boot process with **UEFI** (Unified Extensible Firmware Interface). UEFI has several advantages over BIOS, including better support for 64-bit systems and larger disks. Perhaps the greatest advantage is that UEFI is a single, complete boot manager and therefore is faster than the multistage BIOS boot process.

Whether booting from BIOS or UEFI, the bootstrap program can perform a variety of tasks. In addition to loading the file containing the kernel program into memory, it also runs diagnostics to determine the state of the machine—for example, inspecting memory and the CPU and discovering devices. If the diagnostics pass, the program can continue with the booting steps. The bootstrap can also initialize all aspects of the system, from CPU registers to device controllers and the contents of main memory. Sooner or later, it starts the operating system and mounts the root file system. It is only at this point is the system said to be **running**.

GRUB is an open-source bootstrap program for Linux and UNIX systems. Boot parameters for the system are set in a GRUB configuration file, which is loaded at startup. GRUB is flexible and allows changes to be made at boot time, including modifying kernel parameters and even selecting among different kernels that can be booted. As an example, the following are kernel parameters from the special Linux file `/proc/cmdline`, which is used at boot time:

```
BOOT_IMAGE=/boot/vmlinuz-4.4.0-59-generic  
root=UUID=5f2e2232-4e47-4fe8-ae94-45ea749a5c92
```

`BOOT_IMAGE` is the name of the kernel image to be loaded into memory, and `root` specifies a unique identifier of the root file system.

To save space as well as decrease boot time, the Linux kernel image is a compressed file that is extracted after it is loaded into memory. During the boot process, the boot loader typically creates a temporary RAM file system, known as `initramfs`. This file system contains necessary drivers and kernel modules that must be installed to support the *real* root file system (which is not in main memory). Once the kernel has started and the necessary drivers are installed, the kernel switches the root file system from the temporary RAM location to the appropriate root file system location. Finally, Linux creates the `systemd` process, the initial process in the system, and then starts other services (for example, a web server and/or database). Ultimately, the system will present the user with a login prompt. In Section 11.5.2, we describe the boot process for Windows.

It is worthwhile to note that the booting mechanism is not independent from the boot loader. Therefore, there are specific versions of the GRUB boot loader for BIOS and UEFI, and the firmware must know as well which specific bootloader is to be used.

The boot process for mobile systems is slightly different from that for traditional PCs. For example, although its kernel is Linux-based, Android does not use GRUB and instead leaves it up to vendors to provide boot loaders. The most common Android boot loader is LK (for “little kernel”). Android systems use the same compressed kernel image as Linux, as well as an initial RAM file system. However, whereas Linux discards the `initramfs` once all necessary drivers have been loaded, Android maintains `initramfs` as the root file system for the device. Once the kernel has been loaded and the root file system mounted, Android starts the `init` process and creates a number of services before displaying the home screen.

Finally, boot loaders for most operating systems—including Windows, Linux, and macOS, as well as both iOS and Android—provide booting into **recovery mode** or **single-user mode** for diagnosing hardware issues, fixing corrupt file systems, and even reinstalling the operating system. In addition to hardware failures, computer systems can suffer from software errors and poor operating-system performance, which we consider in the following section.

2.10 Operating-System Debugging

We have mentioned debugging from time to time in this chapter. Here, we take a closer look. Broadly, **debugging** is the activity of finding and fixing errors in a system, both in hardware and in software. Performance problems are considered bugs, so debugging can also include **performance tuning**, which seeks to improve performance by removing processing **bottlenecks**. In this section, we explore debugging process and kernel errors and performance problems. Hardware debugging is outside the scope of this text.

2.10.1 Failure Analysis

If a process fails, most operating systems write the error information to a **log file** to alert system administrators or users that the problem occurred. The operating system can also take a **core dump**—a capture of the memory of the process—and store it in a file for later analysis. (Memory was referred to as the

“core” in the early days of computing.) Running programs and core dumps can be probed by a debugger, which allows a programmer to explore the code and memory of a process at the time of failure.

Debugging user-level process code is a challenge. Operating-system kernel debugging is even more complex because of the size and complexity of the kernel, its control of the hardware, and the lack of user-level debugging tools. A failure in the kernel is called a **crash**. When a crash occurs, error information is saved to a log file, and the memory state is saved to a **crash dump**.

Operating-system debugging and process debugging frequently use different tools and techniques due to the very different nature of these two tasks. Consider that a kernel failure in the file-system code would make it risky for the kernel to try to save its state to a file on the file system before rebooting. A common technique is to save the kernel’s memory state to a section of disk set aside for this purpose that contains no file system. If the kernel detects an unrecoverable error, it writes the entire contents of memory, or at least the kernel-owned parts of the system memory, to the disk area. When the system reboots, a process runs to gather the data from that area and write it to a crash dump file within a file system for analysis. Obviously, such strategies would be unnecessary for debugging ordinary user-level processes.

2.10.2 Performance Monitoring and Tuning

We mentioned earlier that performance tuning seeks to improve performance by removing processing bottlenecks. To identify bottlenecks, we must be able to monitor system performance. Thus, the operating system must have some means of computing and displaying measures of system behavior. Tools may be characterized as providing either *per-process* or *system-wide* observations. To make these observations, tools may use one of two approaches—*counters* or *tracing*. We explore each of these in the following sections.

2.10.2.1 Counters

Operating systems keep track of system activity through a series of counters, such as the number of system calls made or the number of operations performed to a network device or disk. The following are examples of Linux tools that use counters:

Per-Process

- **ps**—reports information for a single process or selection of processes
- **top**—reports real-time statistics for current processes

System-Wide

- **vmstat**—reports memory-usage statistics
- **netstat**—reports statistics for network interfaces
- **iostat**—reports I/O usage for disks

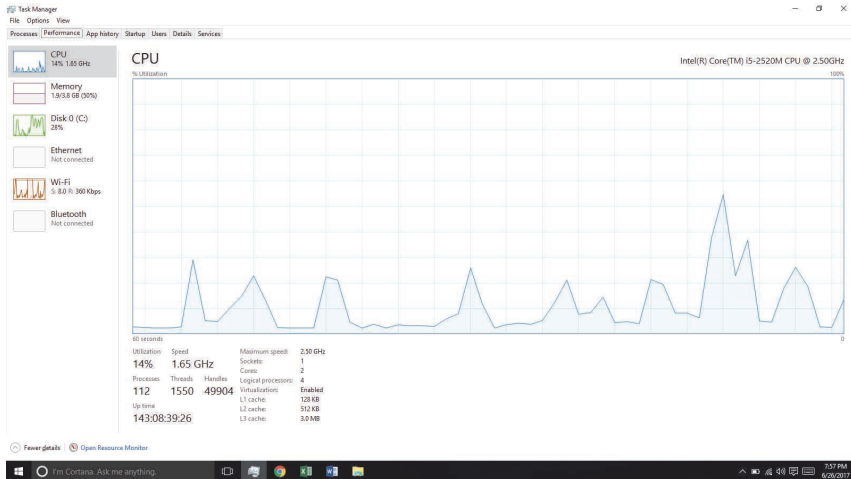


Figure 2.19 The Windows 10 task manager.

Most of the counter-based tools on Linux systems read statistics from the `/proc` file system. `/proc` is a “pseudo” file system that exists only in kernel memory and is used primarily for querying various per-process as well as kernel statistics. The `/proc` file system is organized as a directory hierarchy, with the process (a unique integer value assigned to each process) appearing as a subdirectory below `/proc`. For example, the directory entry `/proc/2155` would contain per-process statistics for the process with an ID of 2155. There are `/proc` entries for various kernel statistics as well. In both this chapter and Chapter 3, we provide programming projects where you will create and access the `/proc` file system.

Windows systems provide the **Windows Task Manager**, a tool that includes information for current applications as well as processes, CPU and memory usage, and networking statistics. A screen shot of the task manager in Windows 10 appears in Figure 2.19.

2.10.3 Tracing

Whereas counter-based tools simply inquire on the current value of certain statistics that are maintained by the kernel, tracing tools collect data for a specific event—such as the steps involved in a system-call invocation.

The following are examples of Linux tools that trace events:

Per-Process

- `strace`—traces system calls invoked by a process
- `gdb`—a source-level debugger

System-Wide

- `perf`—a collection of Linux performance tools
- `tcpdump`—collects network packets

Kernighan's Law

“Debugging is twice as hard as writing the code in the first place. Therefore, if you write the code as cleverly as possible, you are, by definition, not smart enough to debug it.”

Making operating systems easier to understand, debug, and tune as they run is an active area of research and practice. A new generation of kernel-enabled performance analysis tools has made significant improvements in how this goal can be achieved. Next, we discuss BCC, a toolkit for dynamic kernel tracing in Linux.

2.10.4 BCC

Debugging the interactions between user-level and kernel code is nearly impossible without a toolset that understands both sets of code and can instrument their interactions. For that toolset to be truly useful, it must be able to debug any area of a system, including areas that were not written with debugging in mind, and do so without affecting system reliability. This toolset must also have a minimal performance impact—ideally it should have no impact when not in use and a proportional impact during use. The BCC toolkit meets these requirements and provides a dynamic, secure, low-impact debugging environment.

BCC (BPF Compiler Collection) is a rich toolkit that provides tracing features for Linux systems. BCC is a front-end interface to the eBPF (extended Berkeley Packet Filter) tool. The BPF technology was developed in the early 1990s for filtering traffic across a computer network. The “extended” BPF (eBPF) added various features to BPF. eBPF programs are written in a subset of C and are compiled into eBPF instructions, which can be dynamically inserted into a running Linux system. The eBPF instructions can be used to capture specific events (such as a certain system call being invoked) or to monitor system performance (such as the time required to perform disk I/O). To ensure that eBPF instructions are well behaved, they are passed through a **verifier** before being inserted into the running Linux kernel. The verifier checks to make sure that the instructions do not affect system performance or security.

Although eBPF provides a rich set of features for tracing within the Linux kernel, it traditionally has been very difficult to develop programs using its C interface. BCC was developed to make it easier to write tools using eBPF by providing a front-end interface in Python. A BCC tool is written in Python and it embeds C code that interfaces with the eBPF instrumentation, which in turn interfaces with the kernel. The BCC tool also compiles the C program into eBPF instructions and inserts it into the kernel using either probes or tracepoints, two techniques that allow tracing events in the Linux kernel.

The specifics of writing custom BCC tools are beyond the scope of this text, but the BCC package (which is installed on the Linux virtual machine we provide) provides a number of existing tools that monitor several areas

of activity in a running Linux kernel. As an example, the BCC disksnoop tool traces disk I/O activity. Entering the command

```
./disksnoop.py
```

generates the following example output:

TIME(s)	T	BYTES	LAT(ms)
1946.29186700	R	8	0.27
1946.33965000	R	8	0.26
1948.34585000	W	8192	0.96
1950.43251000	R	4096	0.56
1951.74121000	R	4096	0.35

This output tells us the timestamp when the I/O operation occurred, whether the I/O was a Read or Write operation, and how many bytes were involved in the I/O. The final column reflects the duration (expressed as latency or LAT) in milliseconds of the I/O.

Many of the tools provided by BCC can be used for specific applications, such as MySQL databases, as well as Java and Python programs. Probes can also be placed to monitor the activity of a specific process. For example, the command

```
./opensnoop -p 1225
```

will trace `open()` system calls performed only by the process with an identifier of 1225.

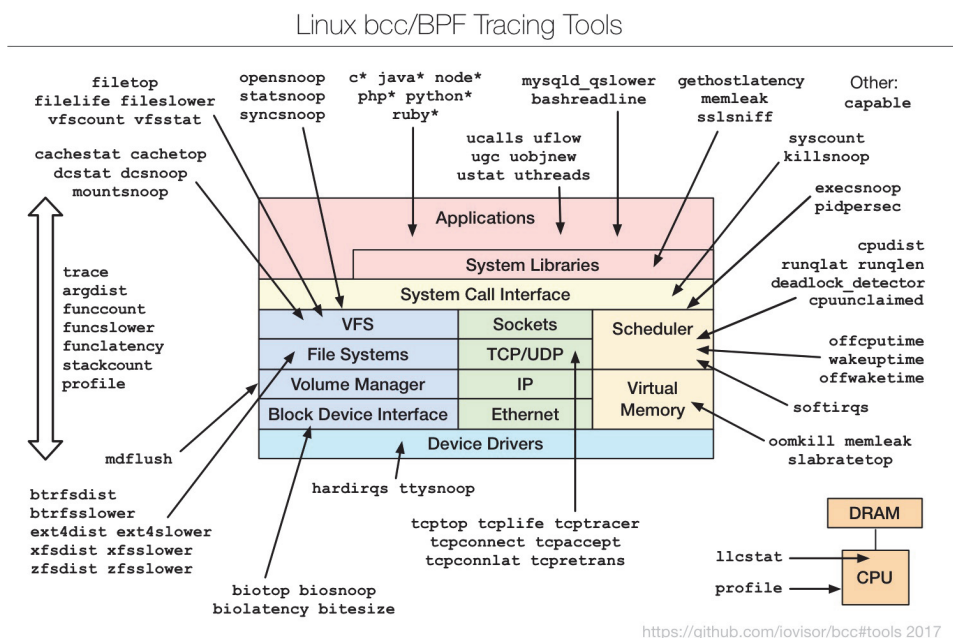


Figure 2.20 The BCC and eBPF tracing tools.

What makes BCC especially powerful is that its tools can be used on live production systems that are running critical applications without causing harm to the system. This is particularly useful for system administrators who must monitor system performance to identify possible bottlenecks or security exploits. Figure 2.20 illustrates the wide range of tools currently provided by BCC and eBPF and their ability to trace essentially any area of the Linux operating system. BCC is a rapidly changing technology with new features constantly being added.

2.11 Summary

- An operating system provides an environment for the execution of programs by providing services to users and programs.
- The three primary approaches for interacting with an operating system are (1) command interpreters, (2) graphical user interfaces, and (3) touch-screen interfaces.
- System calls provide an interface to the services made available by an operating system. Programmers use a system call's application programming interface (API) for accessing system-call services.
- System calls can be divided into six major categories: (1) process control, (2) file management, (3) device management, (4) information maintenance, (5) communications, and (6) protection.
- The standard C library provides the system-call interface for UNIX and Linux systems.
- Operating systems also include a collection of system programs that provide utilities to users.
- A linker combines several relocatable object modules into a single binary executable file. A loader loads the executable file into memory, where it becomes eligible to run on an available CPU.
- There are several reasons why applications are operating-system specific. These include different binary formats for program executables, different instruction sets for different CPUs, and system calls that vary from one operating system to another.
- An operating system is designed with specific goals in mind. These goals ultimately determine the operating system's policies. An operating system implements these policies through specific mechanisms.
- A monolithic operating system has no structure; all functionality is provided in a single, static binary file that runs in a single address space. Although such systems are difficult to modify, their primary benefit is efficiency.
- A layered operating system is divided into a number of discrete layers, where the bottom layer is the hardware interface and the highest layer is the user interface. Although layered software systems have had some suc-

cess, this approach is generally not ideal for designing operating systems due to performance problems.

- The microkernel approach for designing operating systems uses a minimal kernel; most services run as user-level applications. Communication takes place via message passing.
- A modular approach for designing operating systems provides operating-system services through modules that can be loaded and removed during run time. Many contemporary operating systems are constructed as hybrid systems using a combination of a monolithic kernel and modules.
- A boot loader loads an operating system into memory, performs initialization, and begins system execution.
- The performance of an operating system can be monitored using either counters or tracing. Counters are a collection of system-wide or per-process statistics, while tracing follows the execution of a program through the operating system.

Practice Exercises

- 2.1 What is the purpose of system calls?
- 2.2 What is the purpose of the command interpreter? Why is it usually separate from the kernel?
- 2.3 What system calls have to be executed by a command interpreter or shell in order to start a new process on a UNIX system?
- 2.4 What is the purpose of system programs?
- 2.5 What is the main advantage of the layered approach to system design? What are the disadvantages of the layered approach?
- 2.6 List five services provided by an operating system, and explain how each creates convenience for users. In which cases would it be impossible for user-level programs to provide these services? Explain your answer.
- 2.7 Why do some systems store the operating system in firmware, while others store it on disk?
- 2.8 How could a system be designed to allow a choice of operating systems from which to boot? What would the bootstrap program need to do?

Further Reading

[Bryant and O'Hallaron (2015)] provide an overview of computer systems, including the role of the linker and loader. [Atlidakis et al. (2016)] discuss POSIX system calls and how they relate to modern operating systems. [Levin (2013)] covers the internals of both macOS and iOS, and [Levin (2015)] describes details of the Android system. Windows 10 internals are covered in [Russinovich et al. (2017)]. BSD UNIX is described in [McKusick et al. (2015)]. [Love (2010)] and

[Mauerer (2008)] thoroughly discuss the Linux kernel. Solaris is fully described in [McDougall and Mauro (2007)].

Linux source code is available at <http://www.kernel.org>. The Ubuntu ISO image is available from <https://www.ubuntu.com/>.

Comprehensive coverage of Linux kernel modules can be found at <http://www.tldp.org/LDP/lkmpg/2.6/lkmpg.pdf>. [Ward (2015)] and <http://www.ibm.com/developerworks/linux/library/l-linuxboot/> describe the Linux boot process using GRUB. Performance tuning—with a focus on Linux and Solaris systems—is covered in [Gregg (2014)]. Details for the BCC toolkit can be found at <https://github.com/iovisor/bcc/#tools>.

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Chapter 2 Exercises

- 2.9 The services and functions provided by an operating system can be divided into two main categories. Briefly describe the two categories, and discuss how they differ.
- 2.10 Describe three general methods for passing parameters to the operating system.
- 2.11 Describe how you could obtain a statistical profile of the amount of time a program spends executing different sections of its code. Discuss the importance of obtaining such a statistical profile.
- 2.12 What are the advantages and disadvantages of using the same system-call interface for manipulating both files and devices?
- 2.13 Would it be possible for the user to develop a new command interpreter using the system-call interface provided by the operating system?
- 2.14 Describe why Android uses ahead-of-time (AOT) rather than just-in-time (JIT) compilation.
- 2.15 What are the two models of interprocess communication? What are the strengths and weaknesses of the two approaches?
- 2.16 Contrast and compare an application programming interface (API) and an application binary interface (ABI).
- 2.17 Why is the separation of mechanism and policy desirable?
- 2.18 It is sometimes difficult to achieve a layered approach if two components of the operating system are dependent on each other. Identify a scenario in which it is unclear how to layer two system components that require tight coupling of their functionalities.
- 2.19 What is the main advantage of the microkernel approach to system design? How do user programs and system services interact in a microkernel architecture? What are the disadvantages of using the microkernel approach?
- 2.20 What are the advantages of using loadable kernel modules?
- 2.21 How are iOS and Android similar? How are they different?
- 2.22 Explain why Java programs running on Android systems do not use the standard Java API and virtual machine.
- 2.23 The experimental Synthesis operating system has an assembler incorporated in the kernel. To optimize system-call performance, the kernel assembles routines within kernel space to minimize the path that the system call must take through the kernel. This approach is the antithesis of the layered approach, in which the path through the kernel is extended to make building the operating system easier. Discuss the pros and cons of the Synthesis approach to kernel design and system-performance optimization.

Programming Problems

- 2.24 In Section 2.3, we described a program that copies the contents of one file to a destination file. This program works by first prompting the user for the name of the source and destination files. Write this program using either the POSIX or Windows API. Be sure to include all necessary error checking, including ensuring that the source file exists.

Once you have correctly designed and tested the program, if you used a system that supports it, run the program using a utility that traces system calls. Linux systems provide the `strace` utility, and macOS systems use the `dtruss` command. (The `dtruss` command, which actually is a front end to `dtrace`, requires admin privileges, so it must be run using `sudo`.) These tools can be used as follows (assume that the name of the executable file is `FileCopy`):

Linux:

```
strace ./FileCopy
```

macOS:

```
sudo dtruss ./FileCopy
```

Since Windows systems do not provide such a tool, you will have to trace through the Windows version of this program using a debugger.

Programming Projects

Introduction to Linux Kernel Modules

In this project, you will learn how to create a kernel module and load it into the Linux kernel. You will then modify the kernel module so that it creates an entry in the `/proc` file system. The project can be completed using the Linux virtual machine that is available with this text. Although you may use any text editor to write these C programs, you will have to use the *terminal* application to compile the programs, and you will have to enter commands on the command line to manage the modules in the kernel.

As you'll discover, the advantage of developing kernel modules is that it is a relatively easy method of interacting with the kernel, thus allowing you to write programs that directly invoke kernel functions. It is important for you to keep in mind that you are indeed writing *kernel code* that directly interacts with the kernel. That normally means that any errors in the code could crash the system! However, since you will be using a virtual machine, any failures will at worst only require rebooting the system.

I. Kernel Modules Overview

The first part of this project involves following a series of steps for creating and inserting a module into the Linux kernel.

You can list all kernel modules that are currently loaded by entering the command

```
lsmod
```

This command will list the current kernel modules in three columns: name, size, and where the module is being used.

```
#include <linux/init.h>
#include <linux/kernel.h>
#include <linux/module.h>

/* This function is called when the module is loaded. */
int simple_init(void)
{
    printk(KERN_INFO "Loading Kernel Module\n");

    return 0;
}

/* This function is called when the module is removed. */
void simple_exit(void)
{
    printk(KERN_INFO "Removing Kernel Module\n");
}

/* Macros for registering module entry and exit points. */
module_init(simple_init);
module_exit(simple_exit);

MODULE_LICENSE("GPL");
MODULE_DESCRIPTION("Simple Module");
MODULE_AUTHOR("SGG");
```

Figure 2.21 Kernel module `simple.c`.

The program in Figure 2.21 (named `simple.c` and available with the source code for this text) illustrates a very basic kernel module that prints appropriate messages when it is loaded and unloaded.

The function `simple_init()` is the **module entry point**, which represents the function that is invoked when the module is loaded into the kernel. Similarly, the `simple_exit()` function is the **module exit point**—the function that is called when the module is removed from the kernel.

The module entry point function must return an integer value, with 0 representing success and any other value representing failure. The module exit point function returns void. Neither the module entry point nor the module exit point is passed any parameters. The two following macros are used for registering the module entry and exit points with the kernel:

```
module_init(simple_init)
```

```
module_exit(simple_exit)
```

Notice in the figure how the module entry and exit point functions make calls to the `printk()` function. `printk()` is the kernel equivalent of `printf()`, but its output is sent to a kernel log buffer whose contents can be read by the `dmesg` command. One difference between `printf()` and `printk()` is that `printk()` allows us to specify a priority flag, whose values are given in the `<linux/printk.h>` include file. In this instance, the priority is `KERN_INFO`, which is defined as an *informational* message.

The final lines—`MODULE_LICENSE()`, `MODULE_DESCRIPTION()`, and `MODULE_AUTHOR()`—represent details regarding the software license, description of the module, and author. For our purposes, we do not require this information, but we include it because it is standard practice in developing kernel modules.

This kernel module `simple.c` is compiled using the Makefile accompanying the source code with this project. To compile the module, enter the following on the command line:

```
make
```

The compilation produces several files. The file `simple.ko` represents the compiled kernel module. The following step illustrates inserting this module into the Linux kernel.

II. Loading and Removing Kernel Modules

Kernel modules are loaded using the `insmod` command, which is run as follows:

```
sudo insmod simple.ko
```

To check whether the module has loaded, enter the `lsmod` command and search for the module `simple`. Recall that the module entry point is invoked when the module is inserted into the kernel. To check the contents of this message in the kernel log buffer, enter the command

```
dmesg
```

You should see the message "Loading Module."

Removing the kernel module involves invoking the `rmmod` command (notice that the `.ko` suffix is unnecessary):

```
sudo rmmod simple
```

Be sure to check with the `dmesg` command to ensure the module has been removed.

Because the kernel log buffer can fill up quickly, it often makes sense to clear the buffer periodically. This can be accomplished as follows:

```
sudo dmesg -c
```

Proceed through the steps described above to create the kernel module and to load and unload the module. Be sure to check the contents of the kernel log buffer using `dmesg` to ensure that you have followed the steps properly.

As kernel modules are running within the kernel, it is possible to obtain values and call functions that are available only in the kernel and not to regular user applications. For example, the Linux include file `<linux/hash.h>` defines several hashing functions for use within the kernel. This file also defines the constant value `GOLDEN_RATIO_PRIME` (which is defined as an unsigned long). This value can be printed out as follows:

```
printk(KERN_INFO "%lu\n", GOLDEN_RATIO_PRIME);
```

As another example, the include file `<linux/gcd.h>` defines the following function

```
unsigned long gcd(unsigned long a, unsigned b);
```

which returns the greatest common divisor of the parameters `a` and `b`.

Once you are able to correctly load and unload your module, complete the following additional steps:

1. Print out the value of `GOLDEN_RATIO_PRIME` in the `simple_init()` function.
2. Print out the greatest common divisor of 3,300 and 24 in the `simple_exit()` function.

As compiler errors are not often helpful when performing kernel development, it is important to compile your program often by running `make` regularly. Be sure to load and remove the kernel module and check the kernel log buffer using `dmesg` to ensure that your changes to `simple.c` are working properly.

In Section 1.4.3, we described the role of the timer as well as the timer interrupt handler. In Linux, the rate at which the timer ticks (the [tick rate](#)) is the value `HZ` defined in `<asm/param.h>`. The value of `HZ` determines the frequency of the timer interrupt, and its value varies by machine type and architecture. For example, if the value of `HZ` is 100, a timer interrupt occurs 100 times per second, or every 10 milliseconds. Additionally, the kernel keeps track of the global variable `jiffies`, which maintains the number of timer interrupts that have occurred since the system was booted. The `jiffies` variable is declared in the file `<linux/jiffies.h>`.

1. Print out the values of `jiffies` and `HZ` in the `simple_init()` function.
2. Print out the value of `jiffies` in the `simple_exit()` function.

Before proceeding to the next set of exercises, consider how you can use the different values of jiffies in `simple_init()` and `simple_exit()` to determine the number of seconds that have elapsed since the time the kernel module was loaded and then removed.

III. The /proc File System

The /proc file system is a “pseudo” file system that exists only in kernel memory and is used primarily for querying various kernel and per-process statistics.

```
#include <linux/init.h>
#include <linux/kernel.h>
#include <linux/module.h>
#include <linux/proc_fs.h>
#include <asm/uaccess.h>

#define BUFFER_SIZE 128
#define PROC_NAME "hello"

ssize_t proc_read(struct file *file, char __user *usr_buf,
    size_t count, loff_t *pos);

static struct file_operations proc_ops = {
    .owner = THIS_MODULE,
    .read = proc_read,
};

/* This function is called when the module is loaded. */
int proc_init(void)
{
    /* creates the /proc/hello entry */
    proc_create(PROC_NAME, 0666, NULL, &proc_ops);

    return 0;
}

/* This function is called when the module is removed. */
void proc_exit(void)
{
    /* removes the /proc/hello entry */
    remove_proc_entry(PROC_NAME, NULL);
}
```

Figure 2.22 The /proc file-system kernel module, Part 1

This exercise involves designing kernel modules that create additional entries in the /proc file system involving both kernel statistics and information related

to specific processes. The entire program is included in Figure 2.22 and Figure 2.23.

We begin by describing how to create a new entry in the `/proc` file system. The following program example (named `hello.c` and available with the source code for this text) creates a `/proc` entry named `/proc/hello`. If a user enters the command

```
cat /proc/hello
```

the infamous `Hello World` message is returned.

```
/* This function is called each time /proc/hello is read */
ssize_t proc_read(struct file *file, char __user *usr_buf,
    size_t count, loff_t *pos)
{
    int rv = 0;
    char buffer[BUFFER_SIZE];
    static int completed = 0;

    if (completed) {
        completed = 0;
        return 0;
    }

    completed = 1;

    rv = sprintf(buffer, "Hello World\n");

    /* copies kernel space buffer to user space usr_buf */
    copy_to_user(usr_buf, buffer, rv);

    return rv;
}
module_init(proc_init);
module_exit(proc_exit);

MODULE_LICENSE("GPL");
MODULE_DESCRIPTION("Hello Module");
MODULE_AUTHOR("SGG");
```

Figure 2.23 The `/proc` file system kernel module, Part 2

In the module entry point `proc_init()`, we create the new `/proc/hello` entry using the `proc_create()` function. This function is passed `proc_ops`, which contains a reference to a `struct file_operations`. This struct initial-

izes the `.owner` and `.read` members. The value of `.read` is the name of the function `proc_read()` that is to be called whenever `/proc/hello` is read.

Examining this `proc_read()` function, we see that the string "Hello World\n" is written to the variable `buffer` where `buffer` exists in kernel memory. Since `/proc/hello` can be accessed from user space, we must copy the contents of `buffer` to user space using the kernel function `copy_to_user()`. This function copies the contents of kernel memory `buffer` to the variable `usr_buf`, which exists in user space.

Each time the `/proc/hello` file is read, the `proc_read()` function is called repeatedly until it returns 0, so there must be logic to ensure that this function returns 0 once it has collected the data (in this case, the string "Hello World\n") that is to go into the corresponding `/proc/hello` file.

Finally, notice that the `/proc/hello` file is removed in the module exit point `proc_exit()` using the function `remove_proc_entry()`.

IV. Assignment

This assignment will involve designing two kernel modules:

1. Design a kernel module that creates a `/proc` file named `/proc/jiffies` that reports the current value of `jiffies` when the `/proc/jiffies` file is read, such as with the command

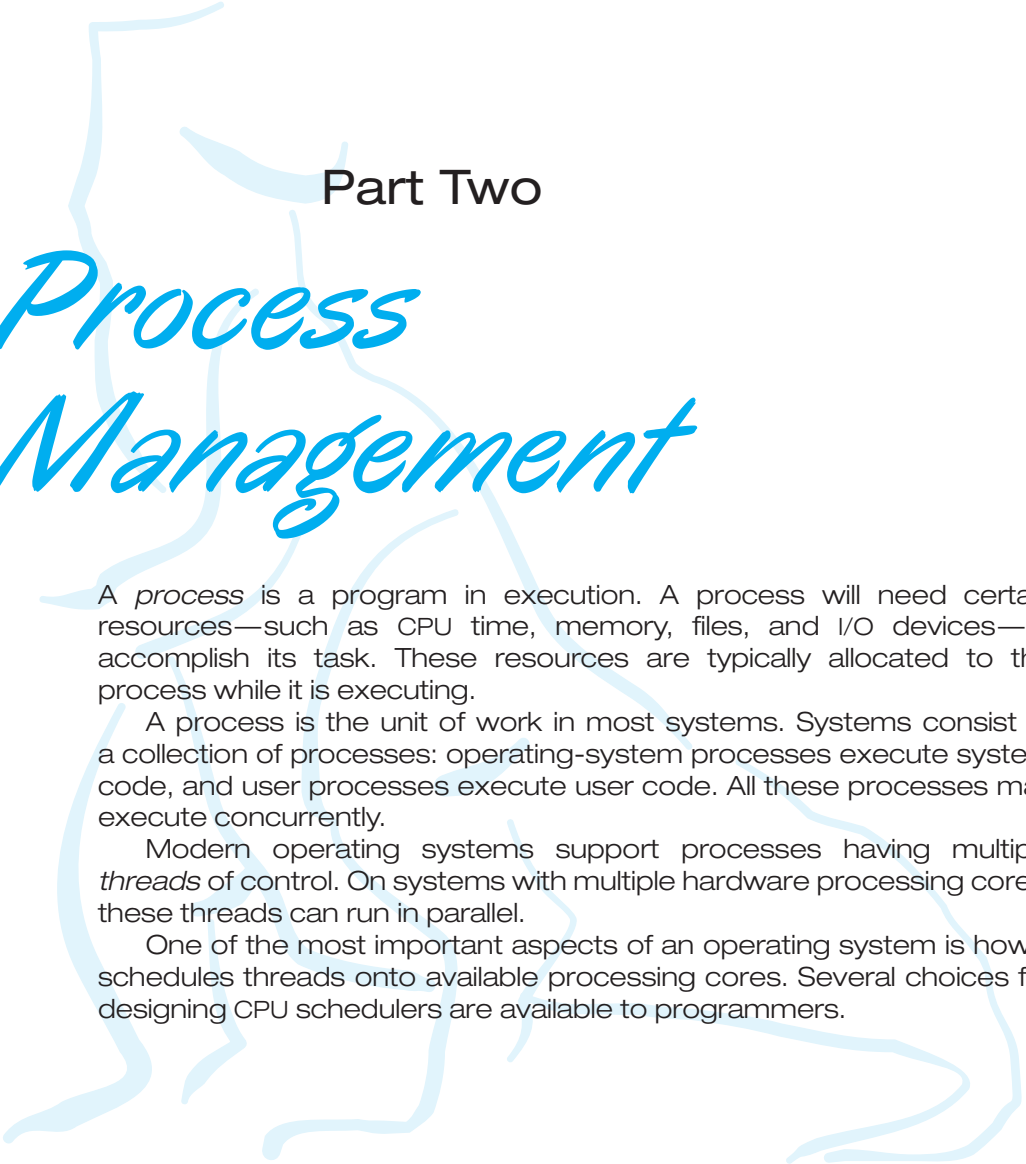
```
cat /proc/jiffies
```

Be sure to remove `/proc/jiffies` when the module is removed.

2. Design a kernel module that creates a `proc` file named `/proc/seconds` that reports the number of elapsed seconds since the kernel module was loaded. This will involve using the value of `jiffies` as well as the HZ rate. When a user enters the command

```
cat /proc/seconds
```

your kernel module will report the number of seconds that have elapsed since the kernel module was first loaded. Be sure to remove `/proc/seconds` when the module is removed.



Part Two

Process Management

A *process* is a program in execution. A process will need certain resources—such as CPU time, memory, files, and I/O devices—to accomplish its task. These resources are typically allocated to the process while it is executing.

A process is the unit of work in most systems. Systems consist of a collection of processes: operating-system processes execute system code, and user processes execute user code. All these processes may execute concurrently.

Modern operating systems support processes having multiple *threads* of control. On systems with multiple hardware processing cores, these threads can run in parallel.

One of the most important aspects of an operating system is how it schedules threads onto available processing cores. Several choices for designing CPU schedulers are available to programmers.

Processes



Early computers allowed only one program to be executed at a time. This program had complete control of the system and had access to all the system's resources. In contrast, contemporary computer systems allow multiple programs to be loaded into memory and executed concurrently. This evolution required firmer control and more compartmentalization of the various programs; and these needs resulted in the notion of a **process**, which is a program in execution. A process is the unit of work in a modern computing system.

The more complex the operating system is, the more it is expected to do on behalf of its users. Although its main concern is the execution of user programs, it also needs to take care of various system tasks that are best done in user space, rather than within the kernel. A system therefore consists of a collection of processes, some executing user code, others executing operating system code. Potentially, all these processes can execute concurrently, with the CPU (or CPUs) multiplexed among them. In this chapter, you will read about what processes are, how they are represented in an operating system, and how they work.

CHAPTER OBJECTIVES

- Identify the separate components of a process and illustrate how they are represented and scheduled in an operating system.
- Describe how processes are created and terminated in an operating system, including developing programs using the appropriate system calls that perform these operations.
- Describe and contrast interprocess communication using shared memory and message passing.
- Design programs that use pipes and POSIX shared memory to perform interprocess communication.
- Describe client–server communication using sockets and remote procedure calls.
- Design kernel modules that interact with the Linux operating system.

3.1 Process Concept

A question that arises in discussing operating systems involves what to call all the CPU activities. Early computers were batch systems that executed **jobs**, followed by the emergence of time-shared systems that ran **user programs**, or **tasks**. Even on a single-user system, a user may be able to run several programs at one time: a word processor, a web browser, and an e-mail package. And even if a computer can execute only one program at a time, such as on an embedded device that does not support multitasking, the operating system may need to support its own internal programmed activities, such as memory management. In many respects, all these activities are similar, so we call all of them **processes**.

Although we personally prefer the more contemporary term *process*, the term *job* has historical significance, as much of operating system theory and terminology was developed during a time when the major activity of operating systems was job processing. Therefore, in some appropriate instances we use *job* when describing the role of the operating system. As an example, it would be misleading to avoid the use of commonly accepted terms that include the word *job* (such as *job scheduling*) simply because *process* has superseded *job*.

3.1.1 The Process

Informally, as mentioned earlier, a process is a program in execution. The status of the current activity of a process is represented by the value of the **program counter** and the contents of the processor's registers. The memory layout of a process is typically divided into multiple sections, and is shown in Figure 3.1. These sections include:

- **Text section**—the executable code
- **Data section**—global variables

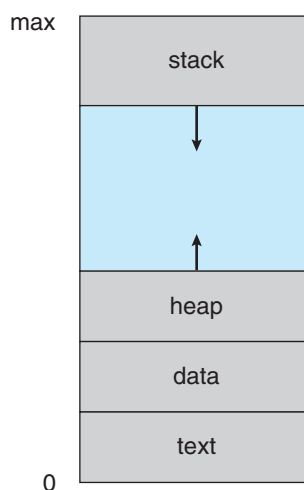


Figure 3.1 Layout of a process in memory.

- **Heap section**—memory that is dynamically allocated during program run time
- **Stack section**—temporary data storage when invoking functions (such as function parameters, return addresses, and local variables)

Notice that the sizes of the text and data sections are fixed, as their sizes do not change during program run time. However, the stack and heap sections can shrink and grow dynamically during program execution. Each time a function is called, an **activation record** containing function parameters, local variables, and the return address is pushed onto the stack; when control is returned from the function, the activation record is popped from the stack. Similarly, the heap will grow as memory is dynamically allocated, and will shrink when memory is returned to the system. Although the stack and heap sections grow *toward* one another, the operating system must ensure they do not *overlap* one another.

We emphasize that a program by itself is not a process. A program is a *passive* entity, such as a file containing a list of instructions stored on disk (often called an **executable file**). In contrast, a process is an *active* entity, with a program counter specifying the next instruction to execute and a set of associated resources. A program becomes a process when an executable file is loaded into memory. Two common techniques for loading executable files are double-clicking an icon representing the executable file and entering the name of the executable file on the command line (as in `prog.exe` or `a.out`).

Although two processes may be associated with the same program, they are nevertheless considered two separate execution sequences. For instance, several users may be running different copies of the mail program, or the same user may invoke many copies of the web browser program. Each of these is a separate process; and although the text sections are equivalent, the data, heap, and stack sections vary. It is also common to have a process that spawns many processes as it runs. We discuss such matters in Section 3.4.

Note that a process can itself be an execution environment for other code. The Java programming environment provides a good example. In most circumstances, an executable Java program is executed within the Java virtual machine (JVM). The JVM executes as a process that interprets the loaded Java code and takes actions (via native machine instructions) on behalf of that code. For example, to run the compiled Java program `Program.class`, we would enter

```
java Program
```

The command `java` runs the JVM as an ordinary process, which in turn executes the Java program `Program` in the virtual machine. The concept is the same as simulation, except that the code, instead of being written for a different instruction set, is written in the Java language.

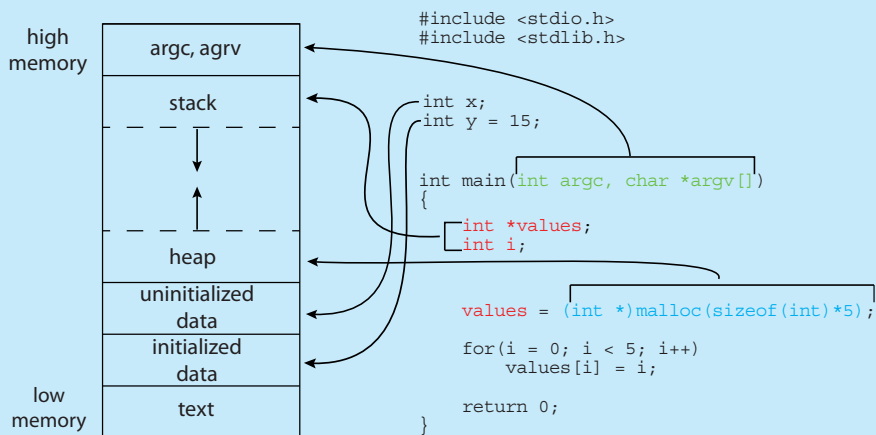
3.1.2 Process State

As a process executes, it changes **state**. The state of a process is defined in part by the current activity of that process. A process may be in one of the following states:

MEMORY LAYOUT OF A C PROGRAM

The figure shown below illustrates the layout of a C program in memory, highlighting how the different sections of a process relate to an actual C program. This figure is similar to the general concept of a process in memory as shown in Figure 3.1, with a few differences:

- The global data section is divided into different sections for (a) initialized data and (b) uninitialized data.
- A separate section is provided for the `argc` and `argv` parameters passed to the `main()` function.



The GNU `size` command can be used to determine the size (in bytes) of some of these sections. Assuming the name of the executable file of the above C program is `memory`, the following is the output generated by entering the command `size memory`:

text	data	bss	dec	hex	filename
1158	284	8	1450	5aa	memory

The `data` field refers to uninitialized data, and `bss` refers to initialized data. (`bss` is a historical term referring to *block started by symbol*.) The `dec` and `hex` values are the sum of the three sections represented in decimal and hexadecimal, respectively.

- **New.** The process is being created.
- **Running.** Instructions are being executed.
- **Waiting.** The process is waiting for some event to occur (such as an I/O completion or reception of a signal).
- **Ready.** The process is waiting to be assigned to a processor.

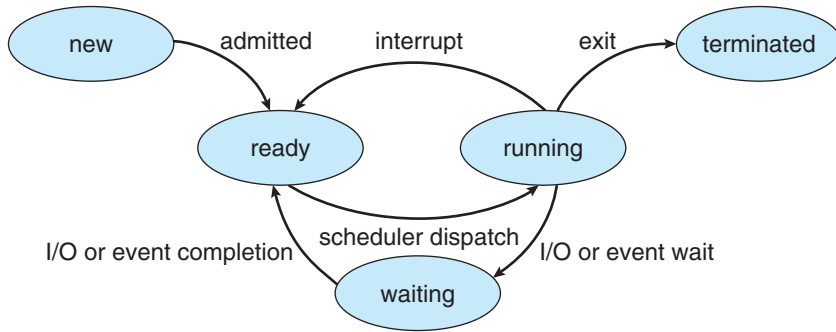


Figure 3.2 Diagram of process state.

- **Terminated.** The process has finished execution.

These names are arbitrary, and they vary across operating systems. The states that they represent are found on all systems, however. Certain operating systems also more finely delineate process states. It is important to realize that only one process can be *running* on any processor core at any instant. Many processes may be *ready* and *waiting*, however. The state diagram corresponding to these states is presented in Figure 3.2.

3.1.3 Process Control Block

Each process is represented in the operating system by a **process control block (PCB)**—also called a **task control block**. A PCB is shown in Figure 3.3. It contains many pieces of information associated with a specific process, including these:

- **Process state.** The state may be new, ready, running, waiting, halted, and so on.
- **Program counter.** The counter indicates the address of the next instruction to be executed for this process.

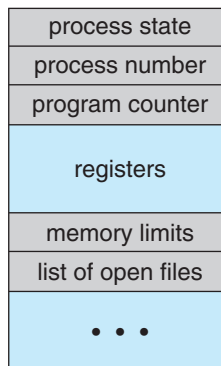


Figure 3.3 Process control block (PCB).

- **CPU registers.** The registers vary in number and type, depending on the computer architecture. They include accumulators, index registers, stack pointers, and general-purpose registers, plus any condition-code information. Along with the program counter, this state information must be saved when an interrupt occurs, to allow the process to be continued correctly afterward when it is rescheduled to run.
- **CPU-scheduling information.** This information includes a process priority, pointers to scheduling queues, and any other scheduling parameters. (Chapter 5 describes process scheduling.)
- **Memory-management information.** This information may include such items as the value of the base and limit registers and the page tables, or the segment tables, depending on the memory system used by the operating system (Chapter 9).
- **Accounting information.** This information includes the amount of CPU and real time used, time limits, account numbers, job or process numbers, and so on.
- **I/O status information.** This information includes the list of I/O devices allocated to the process, a list of open files, and so on.

In brief, the PCB simply serves as the repository for all the data needed to start, or restart, a process, along with some accounting data.

3.1.4 Threads

The process model discussed so far has implied that a process is a program that performs a single **thread** of execution. For example, when a process is running a word-processor program, a single thread of instructions is being executed. This single thread of control allows the process to perform only one task at a time. Thus, the user cannot simultaneously type in characters and run the spell checker. Most modern operating systems have extended the process concept to allow a process to have multiple threads of execution and thus to perform more than one task at a time. This feature is especially beneficial on multicore systems, where multiple threads can run in parallel. A multithreaded word processor could, for example, assign one thread to manage user input while another thread runs the spell checker. On systems that support threads, the PCB is expanded to include information for each thread. Other changes throughout the system are also needed to support threads. Chapter 4 explores threads in detail.

3.2 Process Scheduling

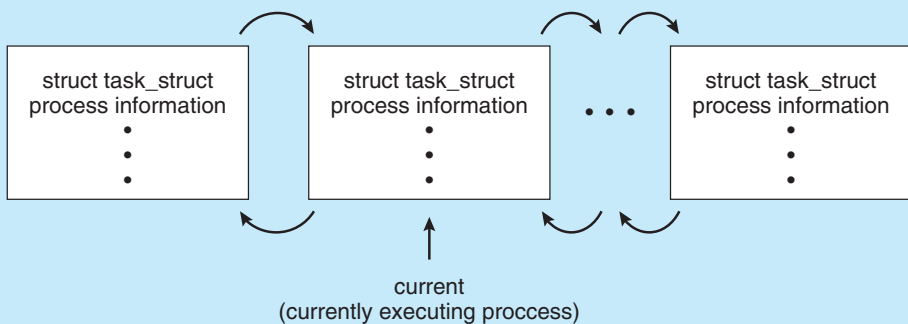
The objective of multiprogramming is to have some process running at all times so as to maximize CPU utilization. The objective of time sharing is to switch a CPU core among processes so frequently that users can interact with each program while it is running. To meet these objectives, the **process scheduler** selects an available process (possibly from a set of several available processes) for program execution on a core. Each CPU core can run one process at a time.

PROCESS REPRESENTATION IN LINUX

The process control block in the Linux operating system is represented by the C structure `task_struct`, which is found in the `<include/linux/sched.h>` include file in the kernel source-code directory. This structure contains all the necessary information for representing a process, including the state of the process, scheduling and memory-management information, list of open files, and pointers to the process's parent and a list of its children and siblings. (A process's **parent** is the process that created it; its **children** are any processes that it creates. Its **siblings** are children with the same parent process.) Some of these fields include:

```
long state;                /* state of the process */
struct sched_entity se;    /* scheduling information */
struct task_struct *parent; /* this process's parent */
struct list_head children; /* this process's children */
struct files_struct *files; /* list of open files */
struct mm_struct *mm;      /* address space */
```

For example, the state of a process is represented by the field `long state` in this structure. Within the Linux kernel, all active processes are represented using a doubly linked list of `task_struct`. The kernel maintains a pointer—`current`—to the process currently executing on the system, as shown below:



As an illustration of how the kernel might manipulate one of the fields in the `task_struct` for a specified process, let's assume the system would like to change the state of the process currently running to the value `new_state`. If `current` is a pointer to the process currently executing, its state is changed with the following:

```
current->state = new_state;
```

For a system with a single CPU core, there will never be more than one process running at a time, whereas a multicore system can run multiple processes at one time. If there are more processes than cores, excess processes will have

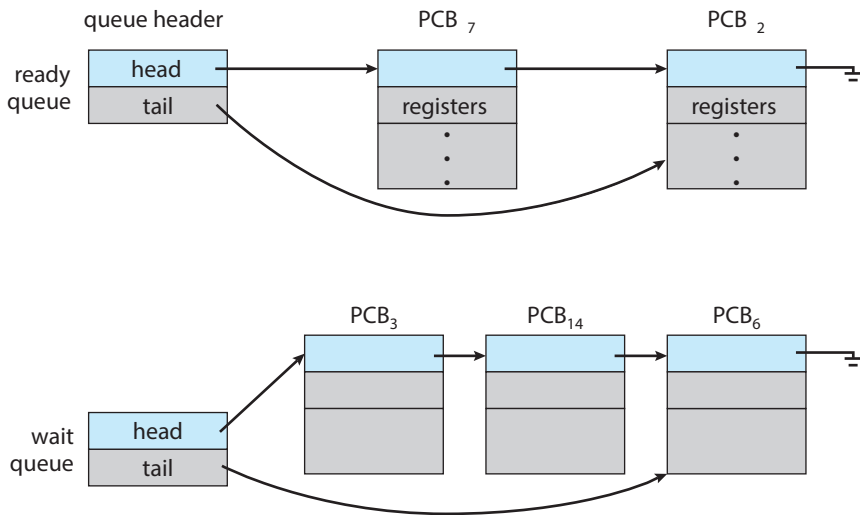


Figure 3.4 The ready queue and wait queues.

to wait until a core is free and can be rescheduled. The number of processes currently in memory is known as the **degree of multiprogramming**.

Balancing the objectives of multiprogramming and time sharing also requires taking the general behavior of a process into account. In general, most processes can be described as either I/O bound or CPU bound. An **I/O-bound process** is one that spends more of its time doing I/O than it spends doing computations. A **CPU-bound process**, in contrast, generates I/O requests infrequently, using more of its time doing computations.

3.2.1 Scheduling Queues

As processes enter the system, they are put into a **ready queue**, where they are ready and waiting to execute on a CPU's core. This queue is generally stored as a linked list; a ready-queue header contains pointers to the first PCB in the list, and each PCB includes a pointer field that points to the next PCB in the ready queue.

The system also includes other queues. When a process is allocated a CPU core, it executes for a while and eventually terminates, is interrupted, or waits for the occurrence of a particular event, such as the completion of an I/O request. Suppose the process makes an I/O request to a device such as a disk. Since devices run significantly slower than processors, the process will have to wait for the I/O to become available. Processes that are waiting for a certain event to occur — such as completion of I/O — are placed in a **wait queue** (Figure 3.4).

A common representation of process scheduling is a **queueing diagram**, such as that in Figure 3.5. Two types of queues are present: the ready queue and a set of wait queues. The circles represent the resources that serve the queues, and the arrows indicate the flow of processes in the system.

A new process is initially put in the ready queue. It waits there until it is selected for execution, or **dispatched**. Once the process is allocated a CPU core and is executing, one of several events could occur:

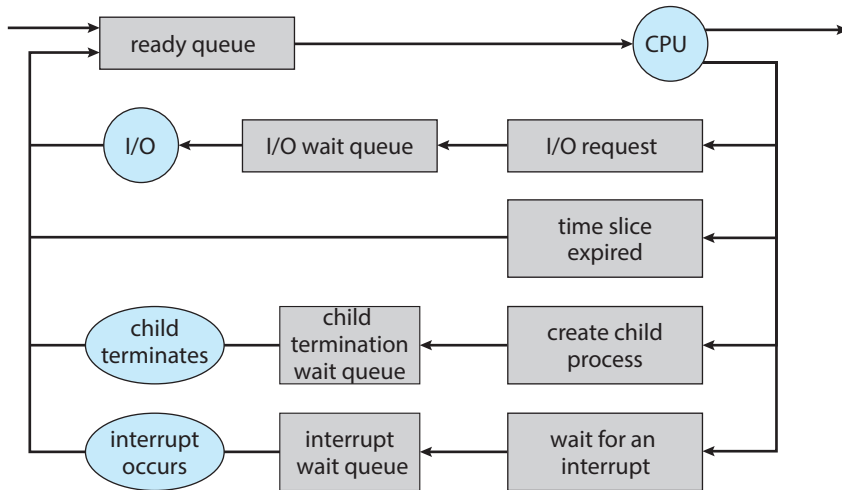


Figure 3.5 Queueing-diagram representation of process scheduling.

- The process could issue an I/O request and then be placed in an I/O wait queue.
- The process could create a new child process and then be placed in a wait queue while it awaits the child's termination.
- The process could be removed forcibly from the core, as a result of an interrupt or having its time slice expire, and be put back in the ready queue.

In the first two cases, the process eventually switches from the waiting state to the ready state and is then put back in the ready queue. A process continues this cycle until it terminates, at which time it is removed from all queues and has its PCB and resources deallocated.

3.2.2 CPU Scheduling

A process migrates among the ready queue and various wait queues throughout its lifetime. The role of the **CPU scheduler** is to select from among the processes that are in the ready queue and allocate a CPU core to one of them. The CPU scheduler must select a new process for the CPU frequently. An I/O-bound process may execute for only a few milliseconds before waiting for an I/O request. Although a CPU-bound process will require a CPU core for longer durations, the scheduler is unlikely to grant the core to a process for an extended period. Instead, it is likely designed to forcibly remove the CPU from a process and schedule another process to run. Therefore, the CPU scheduler executes at least once every 100 milliseconds, although typically much more frequently.

Some operating systems have an intermediate form of scheduling, known as **swapping**, whose key idea is that sometimes it can be advantageous to remove a process from memory (and from active contention for the CPU) and thus reduce the degree of multiprogramming. Later, the process can be reintroduced into memory, and its execution can be continued where it left off. This scheme is known as **swapping** because a process can be “swapped out”

from memory to disk, where its current status is saved, and later “swapped in” from disk back to memory, where its status is restored. Swapping is typically only necessary when memory has been overcommitted and must be freed up. Swapping is discussed in Chapter 9.

3.2.3 Context Switch

As mentioned in Section 1.2.1, interrupts cause the operating system to change a CPU core from its current task and to run a kernel routine. Such operations happen frequently on general-purpose systems. When an interrupt occurs, the system needs to save the current **context** of the process running on the CPU core so that it can restore that context when its processing is done, essentially suspending the process and then resuming it. The context is represented in the PCB of the process. It includes the value of the CPU registers, the process state (see Figure 3.2), and memory-management information. Generically, we perform a **state save** of the current state of the CPU core, be it in kernel or user mode, and then a **state restore** to resume operations.

Switching the CPU core to another process requires performing a state save of the current process and a state restore of a different process. This task is known as a **context switch** and is illustrated in Figure 3.6. When a context switch occurs, the kernel saves the context of the old process in its PCB and loads the saved context of the new process scheduled to run. Context-switch time is pure overhead, because the system does no useful work while switching. Switching speed varies from machine to machine, depending on the

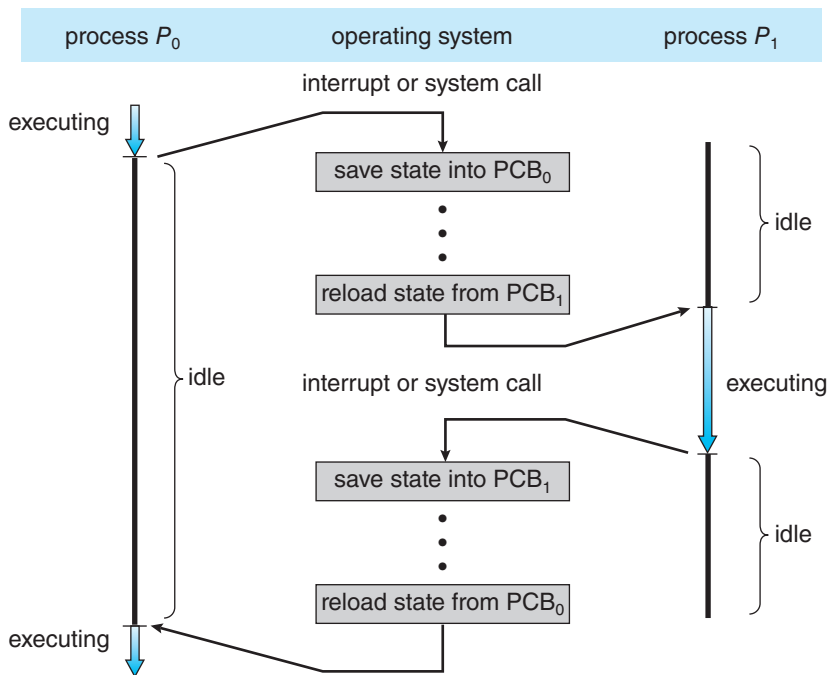


Figure 3.6 Diagram showing context switch from process to process.

MULTITASKING IN MOBILE SYSTEMS

Because of the constraints imposed on mobile devices, early versions of iOS did not provide user-application multitasking; only one application ran in the foreground while all other user applications were suspended. Operating-system tasks were multitasked because they were written by Apple and well behaved. However, beginning with iOS 4, Apple provided a limited form of multitasking for user applications, thus allowing a single foreground application to run concurrently with multiple background applications. (On a mobile device, the **foreground** application is the application currently open and appearing on the display. The **background** application remains in memory, but does not occupy the display screen.) The iOS 4 programming API provided support for multitasking, thus allowing a process to run in the background without being suspended. However, it was limited and only available for a few application types. As hardware for mobile devices began to offer larger memory capacities, multiple processing cores, and greater battery life, subsequent versions of iOS began to support richer functionality for multitasking with fewer restrictions. For example, the larger screen on iPad tablets allowed running two foreground apps at the same time, a technique known as **split-screen**.

Since its origins, Android has supported multitasking and does not place constraints on the types of applications that can run in the background. If an application requires processing while in the background, the application must use a **service**, a separate application component that runs on behalf of the background process. Consider a streaming audio application: if the application moves to the background, the service continues to send audio data to the audio device driver on behalf of the background application. In fact, the service will continue to run even if the background application is suspended. Services do not have a user interface and have a small memory footprint, thus providing an efficient technique for multitasking in a mobile environment.

memory speed, the number of registers that must be copied, and the existence of special instructions (such as a single instruction to load or store all registers). A typical speed is a several microseconds.

Context-switch times are highly dependent on hardware support. For instance, some processors provide multiple sets of registers. A context switch here simply requires changing the pointer to the current register set. Of course, if there are more active processes than there are register sets, the system resorts to copying register data to and from memory, as before. Also, the more complex the operating system, the greater the amount of work that must be done during a context switch. As we will see in Chapter 9, advanced memory-management techniques may require that extra data be switched with each context. For instance, the address space of the current process must be preserved as the space of the next task is prepared for use. How the address space is preserved, and what amount of work is needed to preserve it, depend on the memory-management method of the operating system.

3.3 Operations on Processes

The processes in most systems can execute concurrently, and they may be created and deleted dynamically. Thus, these systems must provide a mechanism for process creation and termination. In this section, we explore the mechanisms involved in creating processes and illustrate process creation on UNIX and Windows systems.

3.3.1 Process Creation

During the course of execution, a process may create several new processes. As mentioned earlier, the creating process is called a parent process, and the new processes are called the children of that process. Each of these new processes may in turn create other processes, forming a **tree** of processes.

Most operating systems (including UNIX, Linux, and Windows) identify processes according to a unique **process identifier** (or **pid**), which is typically an integer number. The pid provides a unique value for each process in the system, and it can be used as an index to access various attributes of a process within the kernel.

Figure 3.7 illustrates a typical process tree for the Linux operating system, showing the name of each process and its pid. (We use the term *process* rather loosely in this situation, as Linux prefers the term *task* instead.) The `systemd` process (which always has a pid of 1) serves as the root parent process for all user processes, and is the first user process created when the system boots. Once the system has booted, the `systemd` process creates processes which provide additional services such as a web or print server, an `ssh` server, and the like. In Figure 3.7, we see two children of `systemd`—`logind` and `sshd`. The `logind` process is responsible for managing clients that directly log onto the system. In this example, a client has logged on and is using the `bash` shell, which has been assigned pid 8416. Using the `bash` command-line interface, this user has created the process `ps` as well as the `vim` editor. The `sshd` process is responsible for managing clients that connect to the system by using `ssh` (which is short for *secure shell*).

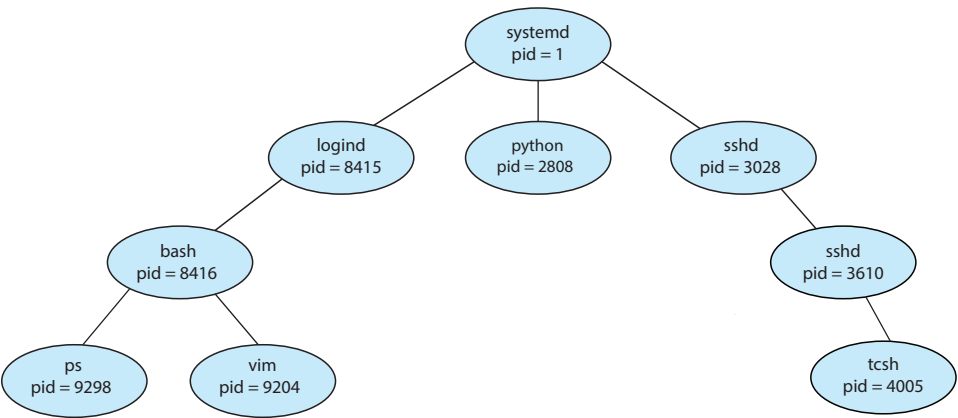


Figure 3.7 A tree of processes on a typical Linux system.

THE `init` AND `systemd` PROCESSES

Traditional UNIX systems identify the process `init` as the root of all child processes. `init` (also known as **System V `init`**) is assigned a pid of 1, and is the first process created when the system is booted. On a process tree similar to what is shown in Figure 3.7, `init` is at the root.

Linux systems initially adopted the System V `init` approach, but recent distributions have replaced it with `systemd`. As described in Section 3.3.1, `systemd` serves as the system's initial process, much the same as System V `init`; however it is much more flexible, and can provide more services, than `init`.

On UNIX and Linux systems, we can obtain a listing of processes by using the `ps` command. For example, the command

```
ps -el
```

will list complete information for all processes currently active in the system. A process tree similar to the one shown in Figure 3.7 can be constructed by recursively tracing parent processes all the way to the `systemd` process. (In addition, Linux systems provide the `pstree` command, which displays a tree of all processes in the system.)

In general, when a process creates a child process, that child process will need certain resources (CPU time, memory, files, I/O devices) to accomplish its task. A child process may be able to obtain its resources directly from the operating system, or it may be constrained to a subset of the resources of the parent process. The parent may have to partition its resources among its children, or it may be able to share some resources (such as memory or files) among several of its children. Restricting a child process to a subset of the parent's resources prevents any process from overloading the system by creating too many child processes.

In addition to supplying various physical and logical resources, the parent process may pass along initialization data (input) to the child process. For example, consider a process whose function is to display the contents of a file—say, `hw1.c`—on the screen of a terminal. When the process is created, it will get, as an input from its parent process, the name of the file `hw1.c`. Using that file name, it will open the file and write the contents out. It may also get the name of the output device. Alternatively, some operating systems pass resources to child processes. On such a system, the new process may get two open files, `hw1.c` and the terminal device, and may simply transfer the datum between the two.

When a process creates a new process, two possibilities for execution exist:

1. The parent continues to execute concurrently with its children.
2. The parent waits until some or all of its children have terminated.

There are also two address-space possibilities for the new process:

1. The child process is a duplicate of the parent process (it has the same program and data as the parent).
2. The child process has a new program loaded into it.

To illustrate these differences, let's first consider the UNIX operating system. In UNIX, as we've seen, each process is identified by its process identifier, which is a unique integer. A new process is created by the `fork()` system call. The new process consists of a copy of the address space of the original process. This mechanism allows the parent process to communicate easily with its child process. Both processes (the parent and the child) continue execution at the instruction after the `fork()`, with one difference: the return code for the `fork()` is zero for the new (child) process, whereas the (nonzero) process identifier of the child is returned to the parent.

After a `fork()` system call, one of the two processes typically uses the `exec()` system call to replace the process's memory space with a new program. The `exec()` system call loads a binary file into memory (destroying the memory image of the program containing the `exec()` system call) and starts

```
#include <sys/types.h>
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    /* fork a child process */
    pid = fork();

    if (pid < 0) { /* error occurred */
        fprintf(stderr, "Fork Failed");
        return 1;
    }
    else if (pid == 0) { /* child process */
        execlp("/bin/ls", "ls", NULL);
    }
    else { /* parent process */
        /* parent will wait for the child to complete */
        wait(NULL);
        printf("Child Complete");
    }

    return 0;
}
```

Figure 3.8 Creating a separate process using the UNIX `fork()` system call.

its execution. In this manner, the two processes are able to communicate and then go their separate ways. The parent can then create more children; or, if it has nothing else to do while the child runs, it can issue a `wait()` system call to move itself off the ready queue until the termination of the child. Because the call to `exec()` overlays the process's address space with a new program, `exec()` does not return control unless an error occurs.

The C program shown in Figure 3.8 illustrates the UNIX system calls previously described. We now have two different processes running copies of the same program. The only difference is that the value of the variable `pid` for the child process is zero, while that for the parent is an integer value greater than zero (in fact, it is the actual `pid` of the child process). The child process inherits privileges and scheduling attributes from the parent, as well certain resources, such as open files. The child process then overlays its address space with the UNIX command `/bin/ls` (used to get a directory listing) using the `execvp()` system call (`execvp()` is a version of the `exec()` system call). The parent waits for the child process to complete with the `wait()` system call. When the child process completes (by either implicitly or explicitly invoking `exit()`), the parent process resumes from the call to `wait()`, where it completes using the `exit()` system call. This is also illustrated in Figure 3.9.

Of course, there is nothing to prevent the child from *not* invoking `exec()` and instead continuing to execute as a copy of the parent process. In this scenario, the parent and child are concurrent processes running the same code instructions. Because the child is a copy of the parent, each process has its own copy of any data.

As an alternative example, we next consider process creation in Windows. Processes are created in the Windows API using the `CreateProcess()` function, which is similar to `fork()` in that a parent creates a new child process. However, whereas `fork()` has the child process inheriting the address space of its parent, `CreateProcess()` requires loading a specified program into the address space of the child process at process creation. Furthermore, whereas `fork()` is passed no parameters, `CreateProcess()` expects no fewer than ten parameters.

The C program shown in Figure 3.10 illustrates the `CreateProcess()` function, which creates a child process that loads the application `mspaint.exe`. We opt for many of the default values of the ten parameters passed to `CreateProcess()`. Readers interested in pursuing the details of process creation and management in the Windows API are encouraged to consult the bibliographical notes at the end of this chapter.

The two parameters passed to the `CreateProcess()` function are instances of the `STARTUPINFO` and `PROCESS_INFORMATION` structures. `STARTUPINFO` specifies many properties of the new process, such as window

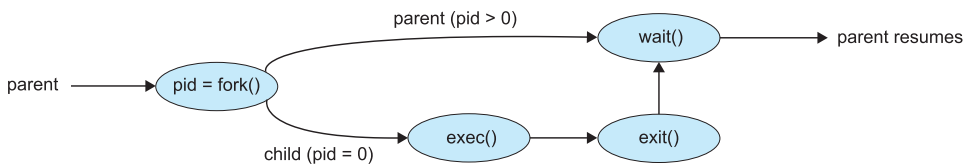


Figure 3.9 Process creation using the `fork()` system call.


```

#include <stdio.h>
#include <windows.h>

int main(VOID)
{
    STARTUPINFO si;
    PROCESS_INFORMATION pi;

    /* allocate memory */
    ZeroMemory(&si, sizeof(si));
    si.cb = sizeof(si);
    ZeroMemory(&pi, sizeof(pi));

    /* create child process */
    if (!CreateProcess(NULL, /* use command line */
        "C:\\WINDOWS\\system32\\mspaint.exe", /* command */
        NULL, /* don't inherit process handle */
        NULL, /* don't inherit thread handle */
        FALSE, /* disable handle inheritance */
        0, /* no creation flags */
        NULL, /* use parent's environment block */
        NULL, /* use parent's existing directory */
        &si,
        &pi))
    {
        fprintf(stderr, "Create Process Failed");
        return -1;
    }
    /* parent will wait for the child to complete */
    WaitForSingleObject(pi.hProcess, INFINITE);
    printf("Child Complete");

    /* close handles */
    CloseHandle(pi.hProcess);
    CloseHandle(pi.hThread);
}

```

Figure 3.10 Creating a separate process using the Windows API.

size and appearance and handles to standard input and output files. The `PROCESS_INFORMATION` structure contains a handle and the identifiers to the newly created process and its thread. We invoke the `ZeroMemory()` function to allocate memory for each of these structures before proceeding with `CreateProcess()`.

The first two parameters passed to `CreateProcess()` are the application name and command-line parameters. If the application name is `NULL` (as it is in this case), the command-line parameter specifies the application to load.

In this instance, we are loading the Microsoft Windows `mspaint.exe` application. Beyond these two initial parameters, we use the default parameters for inheriting process and thread handles as well as specifying that there will be no creation flags. We also use the parent's existing environment block and starting directory. Last, we provide two pointers to the `STARTUPINFO` and `PROCESS_INFORMATION` structures created at the beginning of the program. In Figure 3.8, the parent process waits for the child to complete by invoking the `wait()` system call. The equivalent of this in Windows is `WaitForSingleObject()`, which is passed a handle of the child process—`pi.hProcess`—and waits for this process to complete. Once the child process exits, control returns from the `WaitForSingleObject()` function in the parent process.

3.3.2 Process Termination

A process terminates when it finishes executing its final statement and asks the operating system to delete it by using the `exit()` system call. At that point, the process may return a status value (typically an integer) to its waiting parent process (via the `wait()` system call). All the resources of the process—including physical and virtual memory, open files, and I/O buffers—are deallocated and reclaimed by the operating system.

Termination can occur in other circumstances as well. A process can cause the termination of another process via an appropriate system call (for example, `TerminateProcess()` in Windows). Usually, such a system call can be invoked only by the parent of the process that is to be terminated. Otherwise, a user—or a misbehaving application—could arbitrarily kill another user's processes. Note that a parent needs to know the identities of its children if it is to terminate them. Thus, when one process creates a new process, the identity of the newly created process is passed to the parent.

A parent may terminate the execution of one of its children for a variety of reasons, such as these:

- The child has exceeded its usage of some of the resources that it has been allocated. (To determine whether this has occurred, the parent must have a mechanism to inspect the state of its children.)
- The task assigned to the child is no longer required.
- The parent is exiting, and the operating system does not allow a child to continue if its parent terminates.

Some systems do not allow a child to exist if its parent has terminated. In such systems, if a process terminates (either normally or abnormally), then all its children must also be terminated. This phenomenon, referred to as **cascading termination**, is normally initiated by the operating system.

To illustrate process execution and termination, consider that, in Linux and UNIX systems, we can terminate a process by using the `exit()` system call, providing an exit status as a parameter:

```
/* exit with status 1 */
exit(1);
```

In fact, under normal termination, `exit()` will be called either directly (as shown above) or indirectly, as the C run-time library (which is added to UNIX executable files) will include a call to `exit()` by default.

A parent process may wait for the termination of a child process by using the `wait()` system call. The `wait()` system call is passed a parameter that allows the parent to obtain the exit status of the child. This system call also returns the process identifier of the terminated child so that the parent can tell which of its children has terminated:

```
pid_t pid;
int status;

pid = wait(&status);
```

When a process terminates, its resources are deallocated by the operating system. However, its entry in the process table must remain there until the parent calls `wait()`, because the process table contains the process's exit status. A process that has terminated, but whose parent has not yet called `wait()`, is known as a **zombie** process. All processes transition to this state when they terminate, but generally they exist as zombies only briefly. Once the parent calls `wait()`, the process identifier of the zombie process and its entry in the process table are released.

Now consider what would happen if a parent did not invoke `wait()` and instead terminated, thereby leaving its child processes as **orphans**. Traditional UNIX systems addressed this scenario by assigning the `init` process as the new parent to orphan processes. (Recall from Section 3.3.1 that `init` serves as the root of the process hierarchy in UNIX systems.) The `init` process periodically invokes `wait()`, thereby allowing the exit status of any orphaned process to be collected and releasing the orphan's process identifier and process-table entry.

Although most Linux systems have replaced `init` with `systemd`, the latter process can still serve the same role, although Linux also allows processes other than `systemd` to inherit orphan processes and manage their termination.

3.3.2.1 Android Process Hierarchy

Because of resource constraints such as limited memory, mobile operating systems may have to terminate existing processes to reclaim limited system resources. Rather than terminating an arbitrary process, Android has identified an *importance hierarchy* of processes, and when the system must terminate a process to make resources available for a new, or more important, process, it terminates processes in order of increasing importance. From most to least important, the hierarchy of process classifications is as follows:

- **Foreground process**—The current process visible on the screen, representing the application the user is currently interacting with
- **Visible process**—A process that is not directly visible on the foreground but that is performing an activity that the foreground process is referring to (that is, a process performing an activity whose status is displayed on the foreground process)

- **Service process**—A process that is similar to a background process but is performing an activity that is apparent to the user (such as streaming music)
- **Background process**—A process that may be performing an activity but is not apparent to the user.
- **Empty process**—A process that holds no active components associated with any application

If system resources must be reclaimed, Android will first terminate empty processes, followed by background processes, and so forth. Processes are assigned an importance ranking, and Android attempts to assign a process as high a ranking as possible. For example, if a process is providing a service and is also visible, it will be assigned the more-important visible classification.

Furthermore, Android development practices suggest following the guidelines of the process life cycle. When these guidelines are followed, the state of a process will be saved prior to termination and resumed at its saved state if the user navigates back to the application.

3.4 Interprocess Communication

Processes executing concurrently in the operating system may be either independent processes or cooperating processes. A process is *independent* if it does not share data with any other processes executing in the system. A process is *cooperating* if it can affect or be affected by the other processes executing in the system. Clearly, any process that shares data with other processes is a cooperating process.

There are several reasons for providing an environment that allows process cooperation:

- **Information sharing.** Since several applications may be interested in the same piece of information (for instance, copying and pasting), we must provide an environment to allow concurrent access to such information.
- **Computation speedup.** If we want a particular task to run faster, we must break it into subtasks, each of which will be executing in parallel with the others. Notice that such a speedup can be achieved only if the computer has multiple processing cores.
- **Modularity.** We may want to construct the system in a modular fashion, dividing the system functions into separate processes or threads, as we discussed in Chapter 2.

Cooperating processes require an **interprocess communication (IPC)** mechanism that will allow them to exchange data—that is, send data to and receive data from each other. There are two fundamental models of interprocess communication: **shared memory** and **message passing**. In the shared-memory model, a region of memory that is shared by the cooperating processes is established. Processes can then exchange information by reading and writing data to the shared region. In the message-passing model,

MULTIPROCESS ARCHITECTURE—CHROME BROWSER

Many websites contain active content, such as JavaScript, Flash, and HTML5 to provide a rich and dynamic web-browsing experience. Unfortunately, these web applications may also contain software bugs, which can result in sluggish response times and can even cause the web browser to crash. This isn't a big problem in a web browser that displays content from only one website. But most contemporary web browsers provide tabbed browsing, which allows a single instance of a web browser application to open several websites at the same time, with each site in a separate tab. To switch between the different sites, a user need only click on the appropriate tab. This arrangement is illustrated below:



A problem with this approach is that if a web application in any tab crashes, the entire process—including all other tabs displaying additional websites—crashes as well.

Google's Chrome web browser was designed to address this issue by using a multiprocess architecture. Chrome identifies three different types of processes: browser, renderers, and plug-ins.

- The **browser** process is responsible for managing the user interface as well as disk and network I/O. A new browser process is created when Chrome is started. Only one browser process is created.
- **Renderer** processes contain logic for rendering web pages. Thus, they contain the logic for handling HTML, Javascript, images, and so forth. As a general rule, a new renderer process is created for each website opened in a new tab, and so several renderer processes may be active at the same time.
- A **plug-in** process is created for each type of plug-in (such as Flash or QuickTime) in use. Plug-in processes contain the code for the plug-in as well as additional code that enables the plug-in to communicate with associated renderer processes and the browser process.

The advantage of the multiprocess approach is that websites run in isolation from one another. If one website crashes, only its renderer process is affected; all other processes remain unharmed. Furthermore, renderer processes run in a **sandbox**, which means that access to disk and network I/O is restricted, minimizing the effects of any security exploits.

communication takes place by means of messages exchanged between the cooperating processes. The two communications models are contrasted in Figure 3.11.

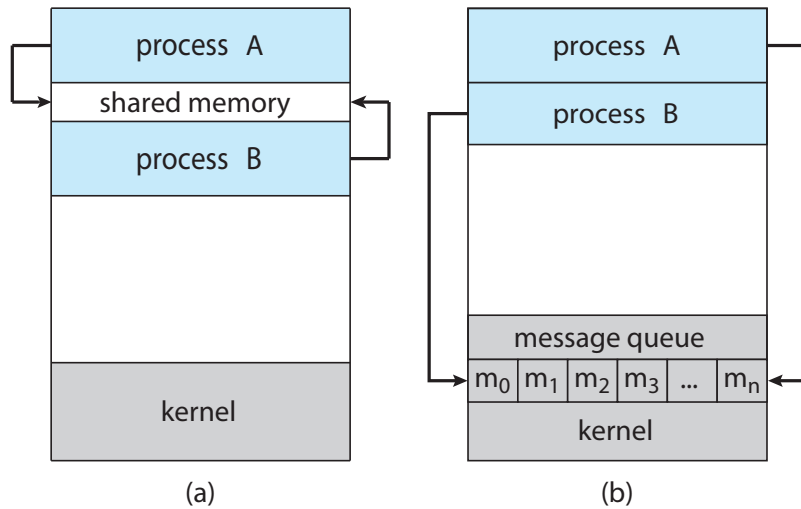


Figure 3.11 Communications models. (a) Shared memory. (b) Message passing.

Both of the models just mentioned are common in operating systems, and many systems implement both. Message passing is useful for exchanging smaller amounts of data, because no conflicts need be avoided. Message passing is also easier to implement in a distributed system than shared memory. (Although there are systems that provide distributed shared memory, we do not consider them in this text.) Shared memory can be faster than message passing, since message-passing systems are typically implemented using system calls and thus require the more time-consuming task of kernel intervention. In shared-memory systems, system calls are required only to establish shared-memory regions. Once shared memory is established, all accesses are treated as routine memory accesses, and no assistance from the kernel is required.

In Section 3.5 and Section 3.6 we explore shared-memory and message-passing systems in more detail.

3.5 IPC in Shared-Memory Systems

Interprocess communication using shared memory requires communicating processes to establish a region of shared memory. Typically, a shared-memory region resides in the address space of the process creating the shared-memory segment. Other processes that wish to communicate using this shared-memory segment must attach it to their address space. Recall that, normally, the operating system tries to prevent one process from accessing another process's memory. Shared memory requires that two or more processes agree to remove this restriction. They can then exchange information by reading and writing data in the shared areas. The form of the data and the location are determined by these processes and are not under the operating system's control. The processes are also responsible for ensuring that they are not writing to the same location simultaneously.

To illustrate the concept of cooperating processes, let's consider the producer–consumer problem, which is a common paradigm for cooperating processes. A **producer** process produces information that is consumed by a **consumer** process. For example, a compiler may produce assembly code that is consumed by an assembler. The assembler, in turn, may produce object modules that are consumed by the loader. The producer–consumer problem also provides a useful metaphor for the client–server paradigm. We generally think of a server as a producer and a client as a consumer. For example, a web server produces (that is, provides) web content such as HTML files and images, which are consumed (that is, read) by the client web browser requesting the resource.

One solution to the producer–consumer problem uses shared memory. To allow producer and consumer processes to run concurrently, we must have available a buffer of items that can be filled by the producer and emptied by the consumer. This buffer will reside in a region of memory that is shared by the producer and consumer processes. A producer can produce one item while the consumer is consuming another item. The producer and consumer must be synchronized, so that the consumer does not try to consume an item that has not yet been produced.

Two types of buffers can be used. The **unbounded buffer** places no practical limit on the size of the buffer. The consumer may have to wait for new items, but the producer can always produce new items. The **bounded buffer** assumes a fixed buffer size. In this case, the consumer must wait if the buffer is empty, and the producer must wait if the buffer is full.

Let's look more closely at how the bounded buffer illustrates interprocess communication using shared memory. The following variables reside in a region of memory shared by the producer and consumer processes:

```
#define BUFFER_SIZE 10

typedef struct {
    . . .
} item;

item buffer[BUFFER_SIZE];
int in = 0;
int out = 0;
```

The shared buffer is implemented as a circular array with two logical pointers: *in* and *out*. The variable *in* points to the next free position in the buffer; *out* points to the first full position in the buffer. The buffer is empty when *in* == *out*; the buffer is full when $((in + 1) \% BUFFER_SIZE) == out$.

The code for the producer process is shown in Figure 3.12, and the code for the consumer process is shown in Figure 3.13. The producer process has a local variable *next_produced* in which the new item to be produced is stored. The consumer process has a local variable *next_consumed* in which the item to be consumed is stored.

This scheme allows at most $BUFFER_SIZE - 1$ items in the buffer at the same time. We leave it as an exercise for you to provide a solution in which $BUFFER_SIZE$ items can be in the buffer at the same time. In Section 3.7.1, we illustrate the POSIX API for shared memory.

```
    item next_produced;

    while (true) {
        /* produce an item in next_produced */

        while (((in + 1) % BUFFER_SIZE) == out)
            ; /* do nothing */

        buffer[in] = next_produced;
        in = (in + 1) % BUFFER_SIZE;
    }
```

Figure 3.12 The producer process using shared memory.

One issue this illustration does not address concerns the situation in which both the producer process and the consumer process attempt to access the shared buffer concurrently. In Chapter 6 and Chapter 7, we discuss how synchronization among cooperating processes can be implemented effectively in a shared-memory environment.

3.6 IPC in Message-Passing Systems

In Section 3.5, we showed how cooperating processes can communicate in a shared-memory environment. The scheme requires that these processes share a region of memory and that the code for accessing and manipulating the shared memory be written explicitly by the application programmer. Another way to achieve the same effect is for the operating system to provide the means for cooperating processes to communicate with each other via a message-passing facility.

```
    item next_consumed;

    while (true) {
        while (in == out)
            ; /* do nothing */

        next_consumed = buffer[out];
        out = (out + 1) % BUFFER_SIZE;

        /* consume the item in next_consumed */
    }
```

Figure 3.13 The consumer process using shared memory.

Message passing provides a mechanism to allow processes to communicate and to synchronize their actions without sharing the same address space. It is particularly useful in a distributed environment, where the communicating processes may reside on different computers connected by a network. For example, an Internet chat program could be designed so that chat participants communicate with one another by exchanging messages.

A message-passing facility provides at least two operations:

```
        send(message)
and
        receive(message)
```

Messages sent by a process can be either fixed or variable in size. If only fixed-sized messages can be sent, the system-level implementation is straightforward. This restriction, however, makes the task of programming more difficult. Conversely, variable-sized messages require a more complex system-level implementation, but the programming task becomes simpler. This is a common kind of tradeoff seen throughout operating-system design.

If processes *P* and *Q* want to communicate, they must send messages to and receive messages from each other: a *communication link* must exist between them. This link can be implemented in a variety of ways. We are concerned here not with the link's physical implementation (such as shared memory, hardware bus, or network, which are covered in Chapter 19) but rather with its logical implementation. Here are several methods for logically implementing a link and the `send()`/`receive()` operations:

- Direct or indirect communication
- Synchronous or asynchronous communication
- Automatic or explicit buffering

We look at issues related to each of these features next.

3.6.1 Naming

Processes that want to communicate must have a way to refer to each other. They can use either direct or indirect communication.

Under **direct communication**, each process that wants to communicate must explicitly name the recipient or sender of the communication. In this scheme, the `send()` and `receive()` primitives are defined as:

- `send(P, message)` — Send a message to process *P*.
- `receive(Q, message)` — Receive a message from process *Q*.

A communication link in this scheme has the following properties:

- A link is established automatically between every pair of processes that want to communicate. The processes need to know only each other's identity to communicate.

- A link is associated with exactly two processes.
- Between each pair of processes, there exists exactly one link.

This scheme exhibits *symmetry* in addressing; that is, both the sender process and the receiver process must name the other to communicate. A variant of this scheme employs *asymmetry* in addressing. Here, only the sender names the recipient; the recipient is not required to name the sender. In this scheme, the `send()` and `receive()` primitives are defined as follows:

- `send(P, message)` — Send a message to process P.
- `receive(id, message)` — Receive a message from any process. The variable `id` is set to the name of the process with which communication has taken place.

The disadvantage in both of these schemes (symmetric and asymmetric) is the limited modularity of the resulting process definitions. Changing the identifier of a process may necessitate examining all other process definitions. All references to the old identifier must be found, so that they can be modified to the new identifier. In general, any such *hard-coding* techniques, where identifiers must be explicitly stated, are less desirable than techniques involving indirection, as described next.

With *indirect communication*, the messages are sent to and received from *mailboxes*, or *ports*. A mailbox can be viewed abstractly as an object into which messages can be placed by processes and from which messages can be removed. Each mailbox has a unique identification. For example, POSIX message queues use an integer value to identify a mailbox. A process can communicate with another process via a number of different mailboxes, but two processes can communicate only if they have a shared mailbox. The `send()` and `receive()` primitives are defined as follows:

- `send(A, message)` — Send a message to mailbox A.
- `receive(A, message)` — Receive a message from mailbox A.

In this scheme, a communication link has the following properties:

- A link is established between a pair of processes only if both members of the pair have a shared mailbox.
- A link may be associated with more than two processes.
- Between each pair of communicating processes, a number of different links may exist, with each link corresponding to one mailbox.

Now suppose that processes P_1 , P_2 , and P_3 all share mailbox A. Process P_1 sends a message to A, while both P_2 and P_3 execute a `receive()` from A. Which process will receive the message sent by P_1 ? The answer depends on which of the following methods we choose:

- Allow a link to be associated with two processes at most.

- Allow at most one process at a time to execute a `receive()` operation.
- Allow the system to select arbitrarily which process will receive the message (that is, either P_2 or P_3 , but not both, will receive the message). The system may define an algorithm for selecting which process will receive the message (for example, *round robin*, where processes take turns receiving messages). The system may identify the receiver to the sender.

A mailbox may be owned either by a process or by the operating system. If the mailbox is owned by a process (that is, the mailbox is part of the address space of the process), then we distinguish between the owner (which can only receive messages through this mailbox) and the user (which can only send messages to the mailbox). Since each mailbox has a unique owner, there can be no confusion about which process should receive a message sent to this mailbox. When a process that owns a mailbox terminates, the mailbox disappears. Any process that subsequently sends a message to this mailbox must be notified that the mailbox no longer exists.

In contrast, a mailbox that is owned by the operating system has an existence of its own. It is independent and is not attached to any particular process. The operating system then must provide a mechanism that allows a process to do the following:

- Create a new mailbox.
- Send and receive messages through the mailbox.
- Delete a mailbox.

The process that creates a new mailbox is that mailbox's owner by default. Initially, the owner is the only process that can receive messages through this mailbox. However, the ownership and receiving privilege may be passed to other processes through appropriate system calls. Of course, this provision could result in multiple receivers for each mailbox.

3.6.2 Synchronization

Communication between processes takes place through calls to `send()` and `receive()` primitives. There are different design options for implementing each primitive. Message passing may be either **blocking** or **nonblocking**—also known as **synchronous** and **asynchronous**. (Throughout this text, you will encounter the concepts of synchronous and asynchronous behavior in relation to various operating-system algorithms.)

- **Blocking send.** The sending process is blocked until the message is received by the receiving process or by the mailbox.
- **Nonblocking send.** The sending process sends the message and resumes operation.
- **Blocking receive.** The receiver blocks until a message is available.
- **Nonblocking receive.** The receiver retrieves either a valid message or a null.

```
message next_produced;

while (true) {
    /* produce an item in next_produced */

    send(next_produced);
}
```

Figure 3.14 The producer process using message passing.

Different combinations of `send()` and `receive()` are possible. When both `send()` and `receive()` are blocking, we have a **rendezvous** between the sender and the receiver. The solution to the producer–consumer problem becomes trivial when we use blocking `send()` and `receive()` statements. The producer merely invokes the blocking `send()` call and waits until the message is delivered to either the receiver or the mailbox. Likewise, when the consumer invokes `receive()`, it blocks until a message is available. This is illustrated in Figures 3.14 and 3.15.

3.6.3 Buffering

Whether communication is direct or indirect, messages exchanged by communicating processes reside in a temporary queue. Basically, such queues can be implemented in three ways:

- **Zero capacity.** The queue has a maximum length of zero; thus, the link cannot have any messages waiting in it. In this case, the sender must block until the recipient receives the message.
- **Bounded capacity.** The queue has finite length n ; thus, at most n messages can reside in it. If the queue is not full when a new message is sent, the message is placed in the queue (either the message is copied or a pointer to the message is kept), and the sender can continue execution without

```
message next_consumed;

while (true) {
    receive(next_consumed);

    /* consume the item in next_consumed */
}
```

Figure 3.15 The consumer process using message passing.

waiting. The link's capacity is finite, however. If the link is full, the sender must block until space is available in the queue.

- **Unbounded capacity.** The queue's length is potentially infinite; thus, any number of messages can wait in it. The sender never blocks.

The zero-capacity case is sometimes referred to as a message system with no buffering. The other cases are referred to as systems with automatic buffering.

3.7 Examples of IPC Systems

In this section, we explore four different IPC systems. We first cover the POSIX API for shared memory and then discuss message passing in the Mach operating system. Next, we present Windows IPC, which interestingly uses shared memory as a mechanism for providing certain types of message passing. We conclude with pipes, one of the earliest IPC mechanisms on UNIX systems.

3.7.1 POSIX Shared Memory

Several IPC mechanisms are available for POSIX systems, including shared memory and message passing. Here, we explore the POSIX API for shared memory.

POSIX shared memory is organized using memory-mapped files, which associate the region of shared memory with a file. A process must first create a shared-memory object using the `shm_open()` system call, as follows:

```
fd = shm_open(name, O_CREAT | O_RDWR, 0666);
```

The first parameter specifies the name of the shared-memory object. Processes that wish to access this shared memory must refer to the object by this name. The subsequent parameters specify that the shared-memory object is to be created if it does not yet exist (`O_CREAT`) and that the object is open for reading and writing (`O_RDWR`). The last parameter establishes the file-access permissions of the shared-memory object. A successful call to `shm_open()` returns an integer file descriptor for the shared-memory object.

Once the object is established, the `ftruncate()` function is used to configure the size of the object in bytes. The call

```
ftruncate(fd, 4096);
```

sets the size of the object to 4,096 bytes.

Finally, the `mmap()` function establishes a memory-mapped file containing the shared-memory object. It also returns a pointer to the memory-mapped file that is used for accessing the shared-memory object.

The programs shown in Figure 3.16 and Figure 3.17 use the producer–consumer model in implementing shared memory. The producer establishes a shared-memory object and writes to shared memory, and the consumer reads from shared memory.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <sys/shm.h>
#include <sys/stat.h>

#include <sys/mman.h>

int main()
{
    /* the size (in bytes) of shared memory object */
    const int SIZE = 4096;
    /* name of the shared memory object */
    const char *name = "OS";
    /* strings written to shared memory */
    const char *message_0 = "Hello";
    const char *message_1 = "World!";

    /* shared memory file descriptor */
    int fd;
    /* pointer to shared memory object */
    char *ptr;

    /* create the shared memory object */
    fd = shm_open(name,O_CREAT | O_RDWR,0666);

    /* configure the size of the shared memory object */
    ftruncate(fd, SIZE);

    /* memory map the shared memory object */
    ptr = (char *)
        mmap(0, SIZE, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0);

    /* write to the shared memory object */
    sprintf(ptr,"%s",message_0);
    ptr += strlen(message_0);
    sprintf(ptr,"%s",message_1);
    ptr += strlen(message_1);

    return 0;
}
```

Figure 3.16 Producer process illustrating POSIX shared-memory API.

The producer, shown in Figure 3.16, creates a shared-memory object named OS and writes the infamous string "Hello World!" to shared memory. The program memory-maps a shared-memory object of the specified size and allows writing to the object. The flag `MAP_SHARED` specifies that changes to the shared-memory object will be visible to all processes sharing the object. Notice that we write to the shared-memory object by calling the `sprintf()` function and writing the formatted string to the pointer `ptr`. After each write, we must increment the pointer by the number of bytes written.

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <sys/shm.h>
#include <sys/stat.h>

#include <sys/mman.h>

int main()
{
    /* the size (in bytes) of shared memory object */
    const int SIZE = 4096;
    /* name of the shared memory object */
    const char *name = "OS";
    /* shared memory file descriptor */
    int fd;
    /* pointer to shared memory object */
    char *ptr;

    /* open the shared memory object */
    fd = shm_open(name, O_RDONLY, 0666);

    /* memory map the shared memory object */
    ptr = (char *)
        mmap(0, SIZE, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0);

    /* read from the shared memory object */
    printf("%s", (char *)ptr);

    /* remove the shared memory object */
    shm_unlink(name);

    return 0;
}
```

Figure 3.17 Consumer process illustrating POSIX shared-memory API.

The consumer process, shown in Figure 3.17, reads and outputs the contents of the shared memory. The consumer also invokes the `shm_unlink()` function, which removes the shared-memory segment after the consumer has accessed it. We provide further exercises using the POSIX shared-memory API in the programming exercises at the end of this chapter. Additionally, we provide more detailed coverage of memory mapping in Section 13.5.

3.7.2 Mach Message Passing

As an example of message passing, we next consider the Mach operating system. Mach was especially designed for distributed systems, but was shown to be suitable for desktop and mobile systems as well, as evidenced by its inclusion in the macOS and iOS operating systems, as discussed in Chapter 2.

The Mach kernel supports the creation and destruction of multiple *tasks*, which are similar to processes but have multiple threads of control and fewer associated resources. Most communication in Mach—including all inter-task communication—is carried out by *messages*. Messages are sent to, and received from, mailboxes, which are called *ports* in Mach. Ports are finite in size and unidirectional; for two-way communication, a message is sent to one port, and a response is sent to a separate *reply* port. Each port may have multiple senders, but only one receiver. Mach uses ports to represent resources such as tasks, threads, memory, and processors, while message passing provides an object-oriented approach for interacting with these system resources and services. Message passing may occur between any two ports on the same host or on separate hosts on a distributed system.

Associated with each port is a collection of *port rights* that identify the capabilities necessary for a task to interact with the port. For example, for a task to receive a message from a port, it must have the capability `MACH_PORT_RIGHT_RECEIVE` for that port. The task that creates a port is that port's owner, and the owner is the only task that is allowed to receive messages from that port. A port's owner may also manipulate the capabilities for a port. This is most commonly done in establishing a reply port. For example, assume that task *T1* owns port *P1*, and it sends a message to port *P2*, which is owned by task *T2*. If *T1* expects to receive a reply from *T2*, it must grant *T2* the right `MACH_PORT_RIGHT_SEND` for port *P1*. Ownership of port rights is at the task level, which means that all threads belonging to the same task share the same port rights. Thus, two threads belonging to the same task can easily communicate by exchanging messages through the per-thread port associated with each thread.

When a task is created, two special ports—the *Task Self* port and the *Notify* port—are also created. The kernel has receive rights to the Task Self port, which allows a task to send messages to the kernel. The kernel can send notification of event occurrences to a task's Notify port (to which, of course, the task has receive rights).

The `mach_port_allocate()` function call creates a new port and allocates space for its queue of messages. It also identifies the rights for the port. Each port right represents a *name* for that port, and a port can only be accessed via

a right. Port names are simple integer values and behave much like UNIX file descriptors. The following example illustrates creating a port using this API:

```
mach_port_t port; // the name of the port right

mach_port_allocate(
    mach_task_self(), // a task referring to itself
    MACH_PORT_RIGHT_RECEIVE, // the right for this port
    &port); // the name of the port right
```

Each task also has access to a **bootstrap port**, which allows a task to register a port it has created with a system-wide **bootstrap server**. Once a port has been registered with the bootstrap server, other tasks can look up the port in this registry and obtain rights for sending messages to the port.

The queue associated with each port is finite in size and is initially empty. As messages are sent to the port, the messages are copied into the queue. All messages are delivered reliably and have the same priority. Mach guarantees that multiple messages from the same sender are queued in first-in, first-out (FIFO) order but does not guarantee an absolute ordering. For instance, messages from two senders may be queued in any order.

Mach messages contain the following two fields:

- A fixed-size message header containing metadata about the message, including the size of the message as well as source and destination ports. Commonly, the sending thread expects a reply, so the port name of the source is passed on to the receiving task, which can use it as a “return address” in sending a reply.
- A variable-sized body containing data.

Messages may be either *simple* or *complex*. A simple message contains ordinary, unstructured user data that are not interpreted by the kernel. A complex message may contain pointers to memory locations containing data (known as “out-of-line” data) or may also be used for transferring port rights to another task. Out-of-line data pointers are especially useful when a message must pass large chunks of data. A simple message would require copying and packaging the data in the message; out-of-line data transmission requires only a pointer that refers to the memory location where the data are stored.

The function `mach_msg()` is the standard API for both sending and receiving messages. The value of one of the function’s parameters—either `MACH_SEND_MSG` or `MACH_RCV_MSG`—indicates if it is a send or receive operation. We now illustrate how it is used when a client task sends a simple message to a server task. Assume there are two ports—`client` and `server`—associated with the client and server tasks, respectively. The code in Figure 3.18 shows the client task constructing a header and sending a message to the server, as well as the server task receiving the message sent from the client.

The `mach_msg()` function call is invoked by user programs for performing message passing. `mach_msg()` then invokes the function `mach_msg_trap()`, which is a system call to the Mach kernel. Within the kernel, `mach_msg_trap()` next calls the function `mach_msg_overwrite_trap()`, which then handles the actual passing of the message.

```

#include<mach/mach.h>

struct message {
    mach_msg_header_t header;
    int data;
};

mach_port_t client;
mach_port_t server;

    /* Client Code */

struct message message;

// construct the header
message.header.msgh_size = sizeof(message);
message.header.msgh_remote_port = server;
message.header.msgh_local_port = client;

// send the message
mach_msg(&message.header, // message header
    MACH_SEND_MSG, // sending a message
    sizeof(message), // size of message sent
    0, // maximum size of received message - unnecessary
    MACH_PORT_NULL, // name of receive port - unnecessary
    MACH_MSG_TIMEOUT_NONE, // no time outs
    MACH_PORT_NULL // no notify port
);

    /* Server Code */

struct message message;

// receive the message
mach_msg(&message.header, // message header
    MACH_RCV_MSG, // sending a message
    0, // size of message sent
    sizeof(message), // maximum size of received message
    server, // name of receive port
    MACH_MSG_TIMEOUT_NONE, // no time outs
    MACH_PORT_NULL // no notify port
);

```

Figure 3.18 Example program illustrating message passing in Mach.

The send and receive operations themselves are flexible. For instance, when a message is sent to a port, its queue may be full. If the queue is not full, the message is copied to the queue, and the sending task continues. If the

port's queue is full, the sender has several options (specified via parameters to `mach_msg()`):

1. Wait indefinitely until there is room in the queue.
2. Wait at most n milliseconds.
3. Do not wait at all but rather return immediately.
4. Temporarily cache a message. Here, a message is given to the operating system to keep, even though the queue to which that message is being sent is full. When the message can be put in the queue, a notification message is sent back to the sender. Only one message to a full queue can be pending at any time for a given sending thread.

The final option is meant for server tasks. After finishing a request, a server task may need to send a one-time reply to the task that requested the service, but it must also continue with other service requests, even if the reply port for a client is full.

The major problem with message systems has generally been poor performance caused by copying of messages from the sender's port to the receiver's port. The Mach message system attempts to avoid copy operations by using virtual-memory-management techniques (Chapter 10). Essentially, Mach maps the address space containing the sender's message into the receiver's address space. Therefore, the message itself is never actually copied, as both the sender and receiver access the same memory. This message-management technique provides a large performance boost but works only for intrasystem messages.

3.7.3 Windows

The Windows operating system is an example of modern design that employs modularity to increase functionality and decrease the time needed to implement new features. Windows provides support for multiple operating environments, or *subsystems*. Application programs communicate with these subsystems via a message-passing mechanism. Thus, application programs can be considered clients of a subsystem server.

The message-passing facility in Windows is called the **advanced local procedure call (ALPC)** facility. It is used for communication between two processes on the same machine. It is similar to the standard remote procedure call (RPC) mechanism that is widely used, but it is optimized for and specific to Windows. (Remote procedure calls are covered in detail in Section 3.8.2.) Like Mach, Windows uses a port object to establish and maintain a connection between two processes. Windows uses two types of ports: **connection ports** and **communication ports**.

Server processes publish connection-port objects that are visible to all processes. When a client wants services from a subsystem, it opens a handle to the server's connection-port object and sends a connection request to that port. The server then creates a channel and returns a handle to the client. The channel consists of a pair of private communication ports: one for client-server messages, the other for server-client messages. Additionally, communication channels support a callback mechanism that allows the client and server to accept requests when they would normally be expecting a reply.

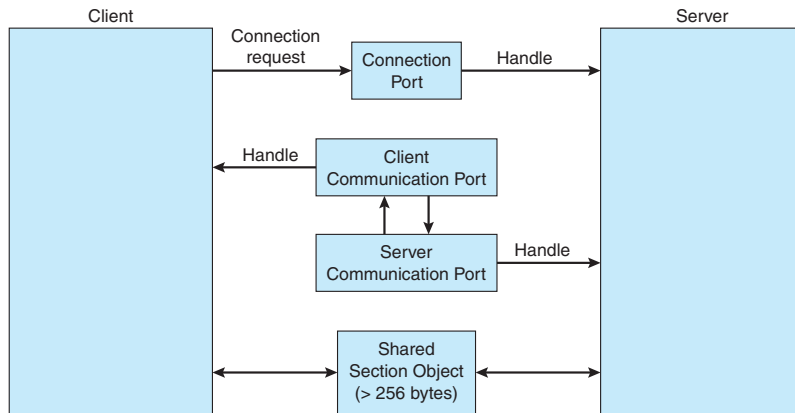


Figure 3.19 Advanced local procedure calls in Windows.

When an ALPC channel is created, one of three message-passing techniques is chosen:

1. For small messages (up to 256 bytes), the port's message queue is used as intermediate storage, and the messages are copied from one process to the other.
2. Larger messages must be passed through a **section object**, which is a region of shared memory associated with the channel.
3. When the amount of data is too large to fit into a section object, an API is available that allows server processes to read and write directly into the address space of a client.

The client has to decide when it sets up the channel whether it will need to send a large message. If the client determines that it does want to send large messages, it asks for a section object to be created. Similarly, if the server decides that replies will be large, it creates a section object. So that the section object can be used, a small message is sent that contains a pointer and size information about the section object. This method is more complicated than the first method listed above, but it avoids data copying. The structure of advanced local procedure calls in Windows is shown in Figure 3.19.

It is important to note that the ALPC facility in Windows is not part of the Windows API and hence is not visible to the application programmer. Rather, applications using the Windows API invoke standard remote procedure calls. When the RPC is being invoked on a process on the same system, the RPC is handled indirectly through an ALPC procedure call. Additionally, many kernel services use ALPC to communicate with client processes.

3.7.4 Pipes

A **pipe** acts as a conduit allowing two processes to communicate. Pipes were one of the first IPC mechanisms in early UNIX systems. They typically provide one of the simpler ways for processes to communicate with one another, although they also have some limitations. In implementing a pipe, four issues must be considered:

1. Does the pipe allow bidirectional communication, or is communication unidirectional?
2. If two-way communication is allowed, is it half duplex (data can travel only one way at a time) or full duplex (data can travel in both directions at the same time)?
3. Must a relationship (such as *parent–child*) exist between the communicating processes?
4. Can the pipes communicate over a network, or must the communicating processes reside on the same machine?

In the following sections, we explore two common types of pipes used on both UNIX and Windows systems: ordinary pipes and named pipes.

3.7.4.1 Ordinary Pipes

Ordinary pipes allow two processes to communicate in standard producer–consumer fashion: the producer writes to one end of the pipe (the **write end**) and the consumer reads from the other end (the **read end**). As a result, ordinary pipes are unidirectional, allowing only one-way communication. If two-way communication is required, two pipes must be used, with each pipe sending data in a different direction. We next illustrate constructing ordinary pipes on both UNIX and Windows systems. In both program examples, one process writes the message *Greetings* to the pipe, while the other process reads this message from the pipe.

On UNIX systems, ordinary pipes are constructed using the function

```
pipe(int fd[])
```

This function creates a pipe that is accessed through the `int fd[]` file descriptors: `fd[0]` is the read end of the pipe, and `fd[1]` is the write end. UNIX treats a pipe as a special type of file. Thus, pipes can be accessed using ordinary `read()` and `write()` system calls.

An ordinary pipe cannot be accessed from outside the process that created it. Typically, a parent process creates a pipe and uses it to communicate with a child process that it creates via `fork()`. Recall from Section 3.3.1 that a child process inherits open files from its parent. Since a pipe is a special type of file, the child inherits the pipe from its parent process. Figure 3.20 illustrates

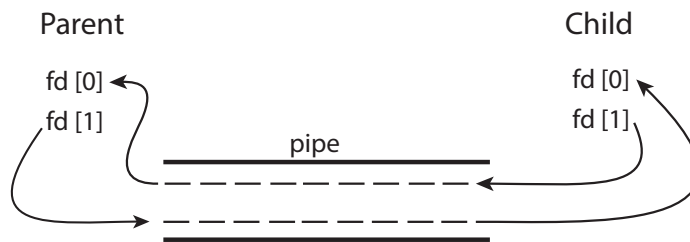


Figure 3.20 File descriptors for an ordinary pipe.

```
#include <sys/types.h>
#include <stdio.h>
#include <string.h>
#include <unistd.h>

#define BUFFER_SIZE 25
#define READ_END 0
#define WRITE_END 1

int main(void)
{
    char write_msg[BUFFER_SIZE] = "Greetings";
    char read_msg[BUFFER_SIZE];
    int fd[2];
    pid_t pid;

    /* Program continues in Figure 3.22 */
```

Figure 3.21 Ordinary pipe in UNIX.

the relationship of the file descriptors in the `fd` array to the parent and child processes. As this illustrates, any writes by the parent to its write end of the pipe—`fd[1]`—can be read by the child from its read end—`fd[0]`—of the pipe.

In the UNIX program shown in Figure 3.21, the parent process creates a pipe and then sends a `fork()` call creating the child process. What occurs after the `fork()` call depends on how the data are to flow through the pipe. In this instance, the parent writes to the pipe, and the child reads from it. It is important to notice that both the parent process and the child process initially close their unused ends of the pipe. Although the program shown in Figure 3.21 does not require this action, it is an important step to ensure that a process reading from the pipe can detect end-of-file (`read()` returns 0) when the writer has closed its end of the pipe.

Ordinary pipes on Windows systems are termed **anonymous pipes**, and they behave similarly to their UNIX counterparts: they are unidirectional and employ parent–child relationships between the communicating processes. In addition, reading and writing to the pipe can be accomplished with the ordinary `ReadFile()` and `WriteFile()` functions. The Windows API for creating pipes is the `CreatePipe()` function, which is passed four parameters. The parameters provide separate handles for (1) reading and (2) writing to the pipe, as well as (3) an instance of the `STARTUPINFO` structure, which is used to specify that the child process is to inherit the handles of the pipe. Furthermore, (4) the size of the pipe (in bytes) may be specified.

Figure 3.23 illustrates a parent process creating an anonymous pipe for communicating with its child. Unlike UNIX systems, in which a child process automatically inherits a pipe created by its parent, Windows requires the programmer to specify which attributes the child process will inherit. This is

```
/* create the pipe */
if (pipe(fd) == -1) {
    fprintf(stderr, "Pipe failed");
    return 1;
}

/* fork a child process */
pid = fork();

if (pid < 0) { /* error occurred */
    fprintf(stderr, "Fork Failed");
    return 1;
}

if (pid > 0) { /* parent process */
    /* close the unused end of the pipe */
    close(fd[READ_END]);

    /* write to the pipe */
    write(fd[WRITE_END], write_msg, strlen(write_msg)+1);

    /* close the write end of the pipe */
    close(fd[WRITE_END]);
}
else { /* child process */
    /* close the unused end of the pipe */
    close(fd[WRITE_END]);

    /* read from the pipe */
    read(fd[READ_END], read_msg, BUFFER_SIZE);
    printf("read %s", read_msg);

    /* close the read end of the pipe */
    close(fd[READ_END]);
}

return 0;
}
```

Figure 3.22 Figure 3.21, continued.

accomplished by first initializing the `SECURITY_ATTRIBUTES` structure to allow handles to be inherited and then redirecting the child process's handles for standard input or standard output to the read or write handle of the pipe. Since the child will be reading from the pipe, the parent must redirect the child's standard input to the read handle of the pipe. Furthermore, as the pipes are half duplex, it is necessary to prohibit the child from inheriting the write end of the